

- 1 Fig. 5 shows part of the curve $y = \operatorname{cosec} x$ together with the x - and y -axes.

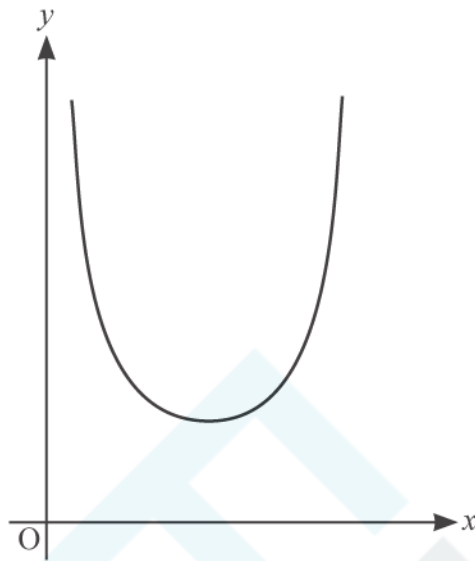


Fig. 5

For the section of the curve which is shown in Fig. 5, write down

- (i) the equations of the two vertical asymptotes,

[2]

- (ii) the coordinates of the minimum point.

[1]

2 In this question you must show detailed reasoning.

Fig. 14 shows the graphs of $y = \sin x \cos 2x$ and $y = \frac{1}{2} - \sin 2x \cos x$.

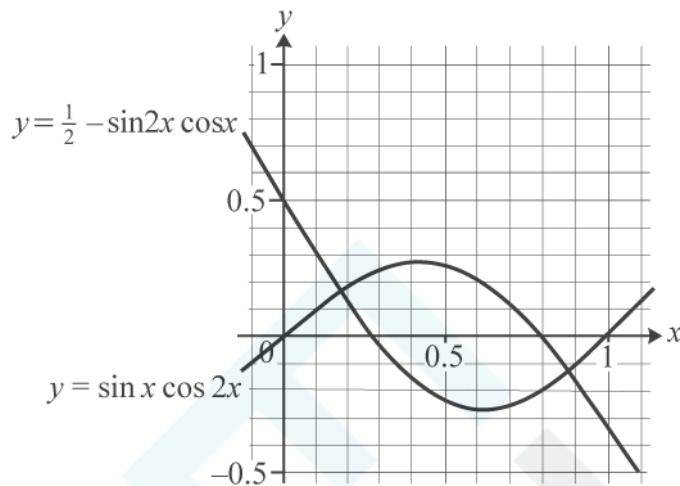


Fig. 14

Use integration to find the area between the two curves, giving your answer in an exact form.

[8]

3(a) In this question you must show detailed reasoning.

The equation of a curve is

$$y = \frac{\sin 2x - x}{x \sin x}.$$

Use the small angle approximation given below and in the Formula Booklet:

$\sin \theta \approx \theta$, $\cos \theta \approx 1 - \frac{1}{2}\theta^2$, $\tan \theta \approx \theta$ where θ is measured in radians to show that

$$\int_{0.01}^{0.05} y \, dx \approx \ln 5.$$

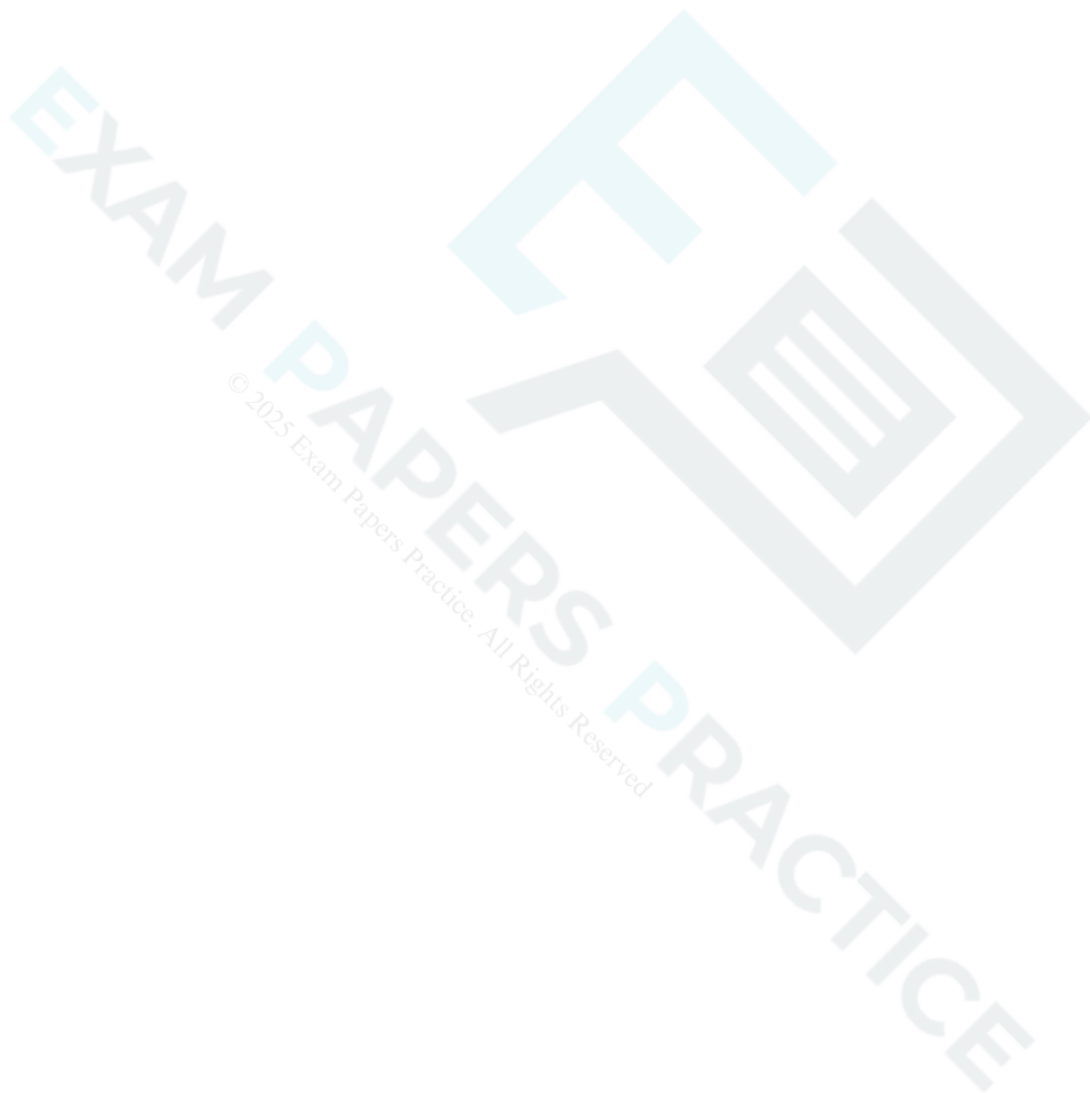
[4]



(b) Use the same small angle approximation to show that

$$\frac{dy}{dx} \approx -10000 \text{ at the point where } x = 0.01.$$

[2]



- (c) The equation $y = 0$ has a root near $x = 1$. Joan uses the Newton-Raphson method to find this root. The output from the spreadsheet she uses is shown in Fig. 10.1.

n	0	1	2	3	4	5	6	7
x_n	1	0.958509	0.950084	0.948261	0.94786	0.947772	0.947753	0.947748

Fig. 10.1

Joan carries out some analysis of this output. The results are shown in Fig. 10.2.

x	y
0.9477475	-7.79967E-07
0.9477485	-2.90821E-06
x	y
0.947745	4.54066E-06
0.947755	-1.67417E-05

Fig. 10.2

Consider the information in Fig. 10.1 and Fig. 10.2.

- Write 4.54066E-06 in standard mathematical notation.
- State the value of the root as accurately as you can, justifying your answer.

[3]

4(a) You are given that $y = 4x + \sin 8x$.

Find $\frac{dy}{dx}$.

[2]

(b)

Find the smallest positive value of x for which $\frac{dy}{dx} = 0$, giving your answer in an exact form.

[2]

5(a) Fig. 2 shows a sector of a circle of radius 8 cm.

The angle of the sector is 2.1 radians.

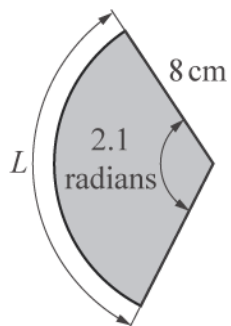


Fig. 2

Calculate the area of the sector.

[2]

(b) Calculate the length of the arc L .

[1]

6 Fig. 1 shows triangle ABC .

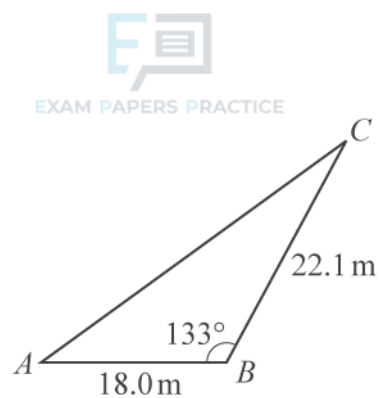


Fig. 1

Calculate the area of triangle ABC , giving your answer correct to 3 significant figures.

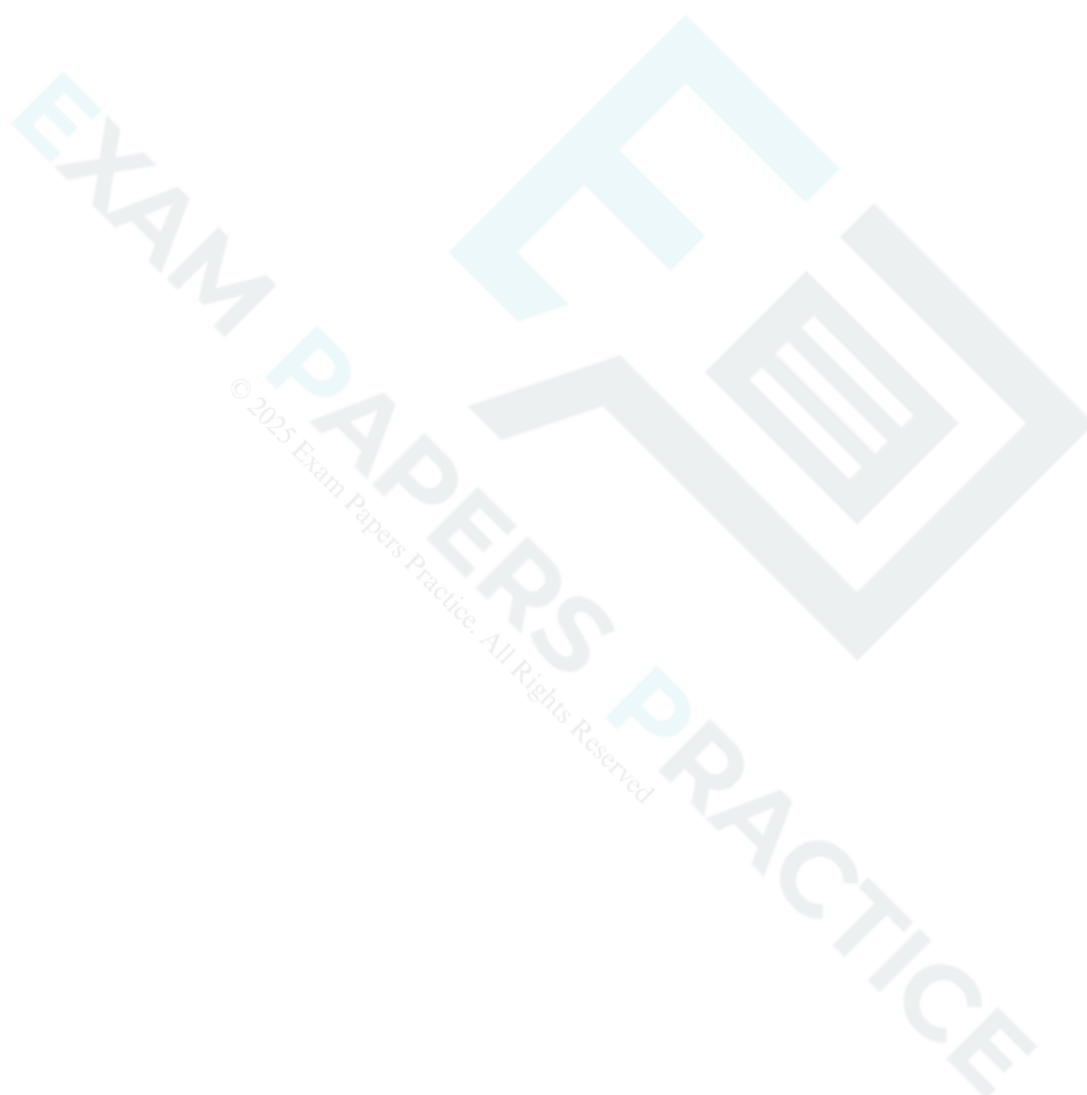
[2]

7 In this question you must show detailed reasoning.

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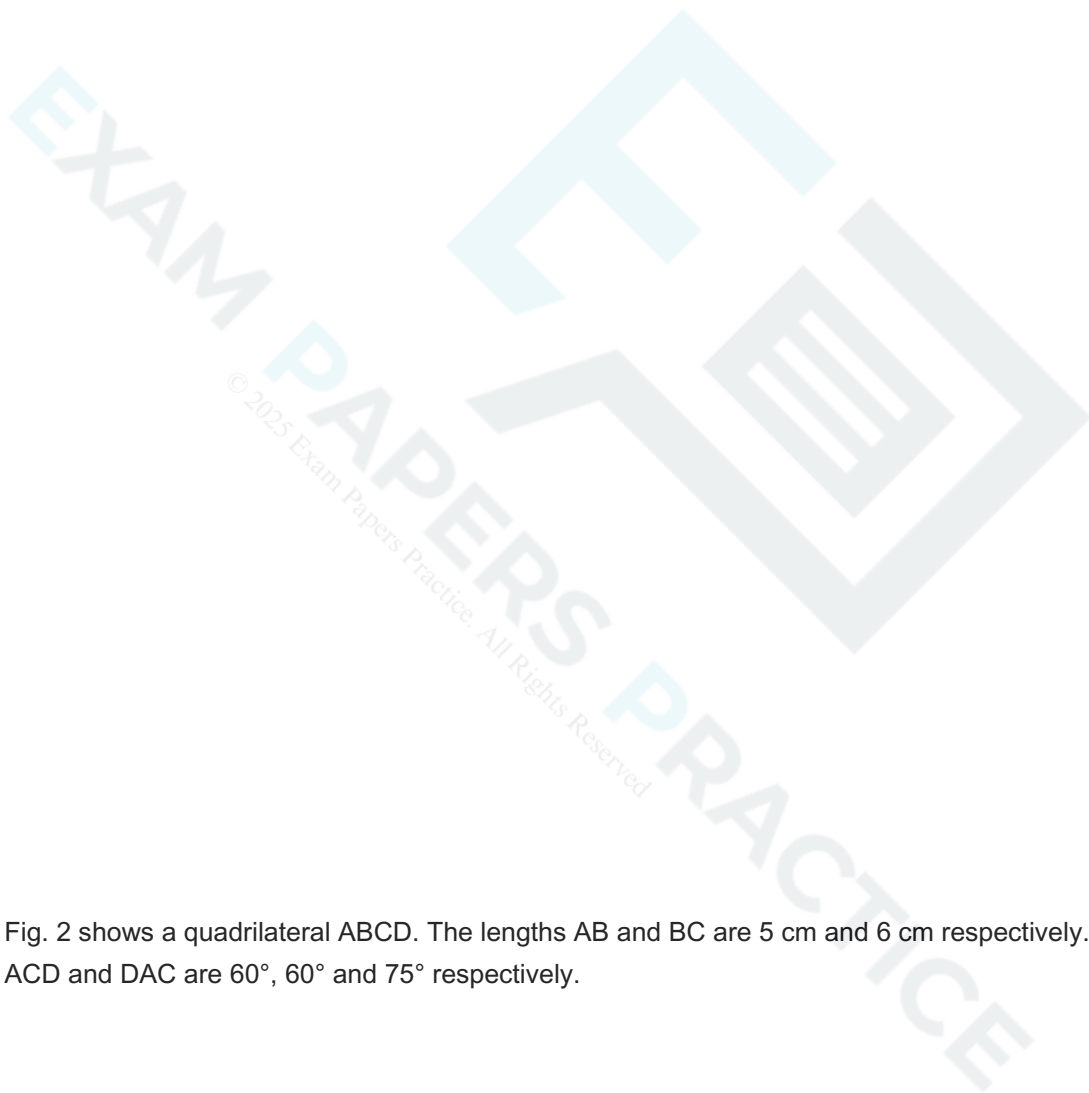
Solve the equation $3 \cos \theta + 8 \tan \theta = 0$ for $0^\circ < \theta < 360^\circ$, giving your answers correct to the nearest degree.

[6]



Use the substitution $x = \tan \theta$ to find the exact value of $\int_{-1}^1 \frac{1}{(1+x^2)^2} dx$.

[8]



- 9 Fig. 2 shows a quadrilateral ABCD. The lengths AB and BC are 5 cm and 6 cm respectively. The angles ABC, ACD and DAC are 60° , 60° and 75° respectively.

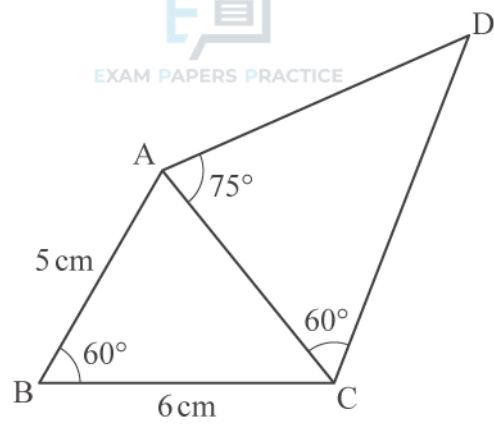


Fig. 2

Calculate the exact value of the length AD .

[4]

10(a) Douglas wants to construct a model for the height of the tide in Liverpool during the day, using a cosine graph to represent the way the height changes.

He knows that the first high tide of the day measures 8.55 m and the first low tide of the day measures 1.75 m.

Douglas uses t for time and h for the height of the tide in metres. With his graph-drawing software set to degrees, he begins by drawing the graph of $h = 5.15 + 3.4 \cos t$.

Verify that this equation gives the correct values of h for the high and low tide.

[1]

(b) Douglas also knows that the first high tide of the day occurs at 1 am and the first low tide occurs at 7.20 am. He wants t to represent the time in hours after midnight, so he modifies his equation to $h = 5.15 + 3.4 \cos(\alpha t + b)$.

(i) Show that Douglas's modified equation gives the first high tide of the day occurring at the correct time if $\alpha + b = 0$. [1]

(ii) Use the time of the first low tide of the day to form a second equation relating α and b . [1]

(iii) Hence show that $\alpha = 28.42$ correct to 2 decimal places. [2]

- (c) Douglas can only sail his boat when the height of the tide is at least 3 m.

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Use the model to predict the range of times that morning when he cannot sail.

[3]

- (d) The next high tide occurs at 12.59 pm when the height of the tide is 8.91 m.

Comment on the suitability of Douglas's model.

[2]

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11(a)

Give details of a sequence of two transformations which maps the curve $y = \sec x$ onto the curve

$$y = \frac{1}{7 \cos x - 2 \sin x}.$$

[3]

(b)

Express $7 \cos x - 2 \sin x$ in the form $R \cos (x + \alpha)$ where $R > 0$ and $0 < \alpha < \frac{1}{2}\pi$, giving the exact value of R and the value of α correct to 3 significant figures.

[4]

12(a) The area of a sector of a circle is 36.288cm^2 . The angle of the sector is θ radians and the radius of the circle is r cm.

Find an expression for θ in terms of r .

[1]

(b) The perimeter of the sector is 24.48cm .

Show that $\theta = \frac{24.48}{r} - 2$.

[1]

(c) Find the possible values of r .

[3]

13(a)

Prove that $\frac{\sin \theta}{1 - \cos \theta} - \frac{1}{\sin \theta} = \cot \theta$.

[4]

(b)

Hence find the exact roots of the equation $\frac{\sin \theta}{1 - \cos \theta} - \frac{1}{\sin \theta} = 3 \tan \theta$ in the interval $0 \leq \theta \leq \pi$.

[3]

14(a) Fig. 14 shows a circle with centre O and radius r cm. The chord AB is such that angle $AOB = x$ radians. The area of the shaded segment formed by AB is 5% of the area of the circle.

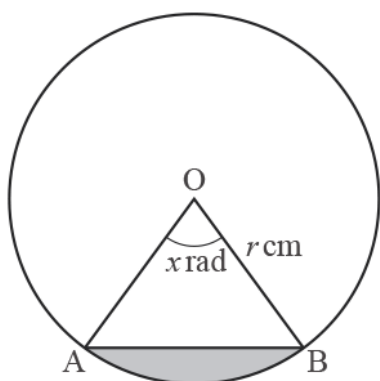


Fig. 14

Show that $x - \sin x - \frac{1}{10}\pi = 0$.

[4]

- (b) The Newton-Raphson method is to be used to find x .

Write down the iterative formula to be used for the equation in part (a).

[1]

- (c) Use three iterations of the Newton-Raphson method with $x_0 = 1.2$ to find the value of x to a suitable degree of accuracy.

[3]

- 15 Fig. 2 shows a triangle with one angle of 117° given. The lengths are given in centimetres.

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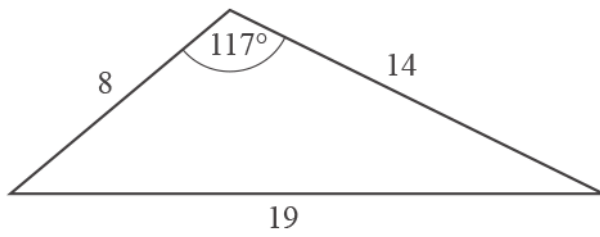


Fig. 2

Calculate the area of the triangle, giving your answer correct to 3 significant figures.

[2]

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16(a) A triangle ABC has sides $AB = 5$ cm, $AC = 9$ cm and $BC = 10$ cm.

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Find the exact area of the triangle.

[3]

(b) Find the cosine of angle BAC, giving your answer as a fraction in its lowest terms.

[2]

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Fig. 2 shows the graphs of $y = 4 \sin x^\circ$ and $y = 3 \cos x^\circ$ for $0 \leq x \leq 360$.

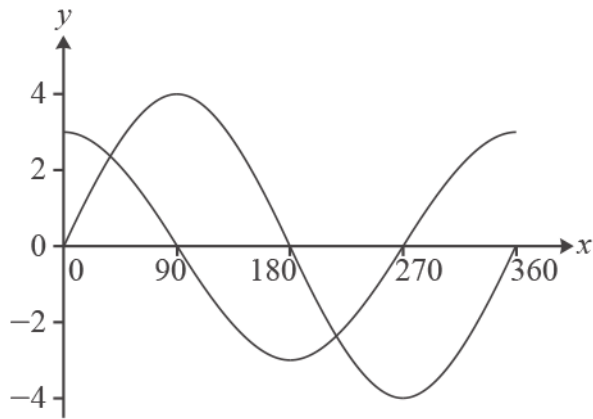


Fig. 2

Find the x-coordinates of the two points of intersection, giving your answers correct to 1 decimal place.

[3]



The function $h(x) = \sin\left(\frac{1}{x}\right)$ is defined for the domain $x > \frac{2}{\pi}$.

(a) Differentiate $h(x)$ with respect to x .

[2]

(b) Find the values between which $\frac{1}{x}$ lies for $x > \frac{2}{\pi}$.

[2]

(c) Show that $h(x)$ is a decreasing function.

[3]

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$$x = 2 \cot \theta, \quad y = 2 \sin^2 \theta,$$

for values of θ for which both x and y are defined.

- (a) For what values of θ , in the interval $0 \leq \theta \leq 2\pi$, is x undefined?

[2]

- (b) Show that the cartesian equation of the curve is $y = \frac{A}{x^2 + B}$ where A and B are positive integers to be determined.

[3]

- (c) Ali says that the curve lies completely above the x -axis. Determine whether Ali is correct.

[2]

- 20 The day length, Y hours, is defined as the difference between the time the sun rises and the time the sun sets on a particular day. For Manchester, England, the following model is proposed for years which are not leap years.

$$Y = a \sin\left(\frac{2\pi}{365}t + b\right) + c,$$

where t is the time in days since the start of the year and a , b and c are constants.

The maximum value of Y , which is 17.03, occurs on June 21st, when $t = 172$. The minimum value of Y , which is 7.47, occurs on December 21st, when $t = 355$.

- (a) Show that $a = 4.78$ and $c = 12.25$.

[2]

- (b) Determine the value of b correct to 3 significant figures.

[2]

On September 1st, when $t = 244$, the day length is recorded as 13.76 hours.

- (c) Show that the model is a good fit for this value.

[2]

In Reykjavik, Iceland, on June 21st the maximum day length was 21.14 hours and on December 21st the minimum day length was 4.12 hours.

- (d) Use this information to refine the model for Manchester to produce a possible model for the day length in Reykjavik.

[1]

On September 1st the day length in Reykjavik is recorded as 14.56 hours.

- (e) Determine whether your possible model for Reykjavik is a good fit for this value.

[1]

21 (a) Write $\cos^2 x$ in terms of $\cos 2x$.

[1]

(b) Express $6 \sin 2x + 8 \cos 2x$ in the form $R \cos(2x - \theta)$, where $0 < \theta < \frac{\pi}{2}$.

[2]

In this question you must show detailed reasoning.

(c) Hence solve the equation $6 \sin 2x + 16 \cos^2 x = 13$ for $0 \leq x \leq 2\pi$ giving your answers correct to 3 significant figures.

[5]

22 Chione uses the equation $V = 22000 - a\sqrt{t}$ to model the value of her caravan, where V is the value in pounds t years after purchase.

(a) The model must give the value of the caravan after 4 years as £12 000. Find the value of a .

[1]

(b) Find the rate at which the value is changing when the caravan is 4 years old.

[3]

(c) Explain the limitations of this model for large values of t .

[1]

Chione creates a second model $V = be^{-0.15t} + c$ in which the initial value is £22 000 and the value after a long time tends to £1000.

(d) Find the values of the constants b and c .

[2]

Chione wishes to find the times other than $t = 0$ at which the two models give the same value.

(e) Show that these times satisfy the equation $t = -\frac{20}{3} \ln\left(1 - \frac{5}{21}\sqrt{t}\right)$.

[3]

Chione uses fixed point iteration with this equation and $t_0 = 5$. The table shows some of her values.

t_0	5
t_1	
t_2	
t_3	5.157 893 470
t_4	5.187 562 897
t_5	5.210 144 461

(f) Find the missing values t_1 and t_2 .

[2]

Fig. 14 shows the graphs of $y = t$ and $y = -\frac{20}{3} \ln\left(1 - \frac{5}{21}\sqrt{t}\right)$.

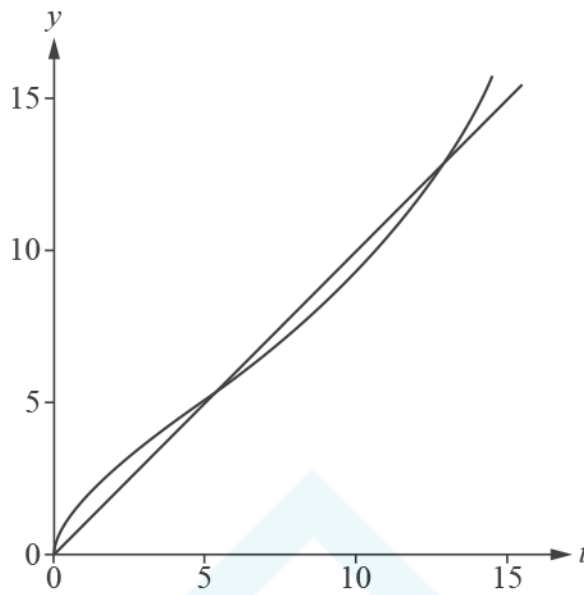
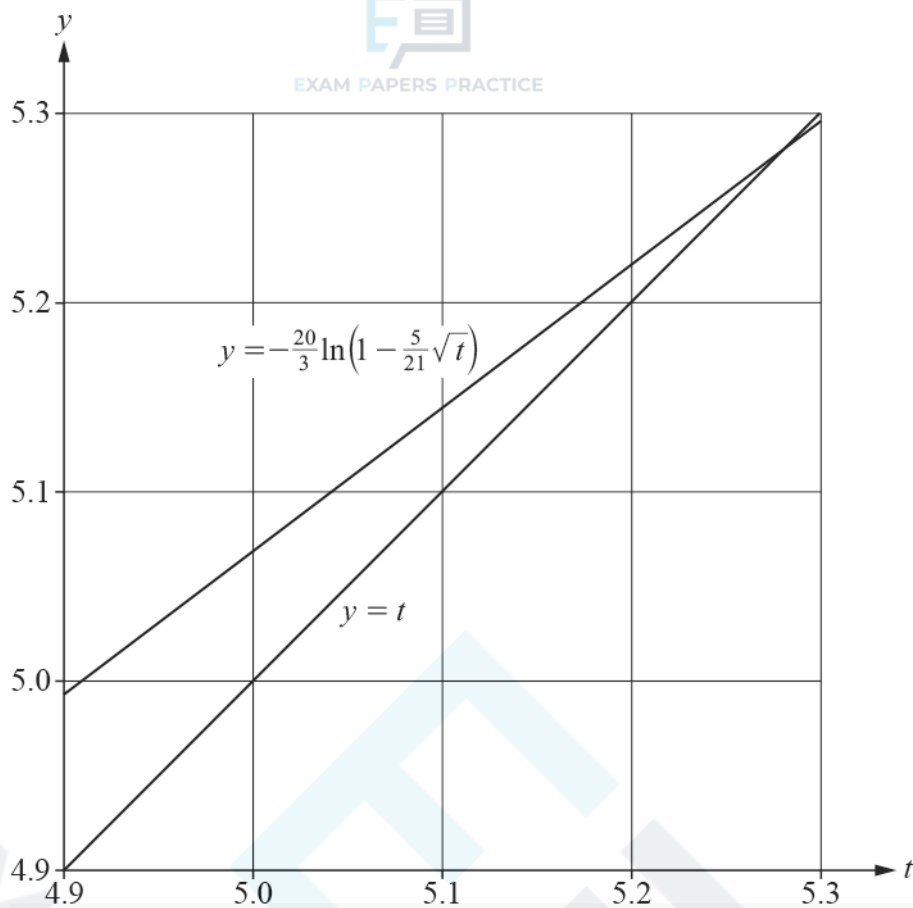


Fig. 14

(g) Sketch the staircase diagram for Chione's iteration on the diagram below.

[1]



- (h) Chione notices that t_4 and t_5 both round to 5.2 so argues that the root of the equation is 5.2 correct to 1 decimal place. Comment on the validity of her argument.

[1]

- (i) Determine whether this fixed point iteration can be used to find the root near to 12.

[2]



The function $f(x)$ for the domain $-\frac{1}{2}\pi \leq x \leq \frac{1}{2}\pi$ is defined by $f(x) = e^{kx} \sin 2x$, where k is a non-zero constant. Fig. 13 shows the graph of $y = f(x)$.

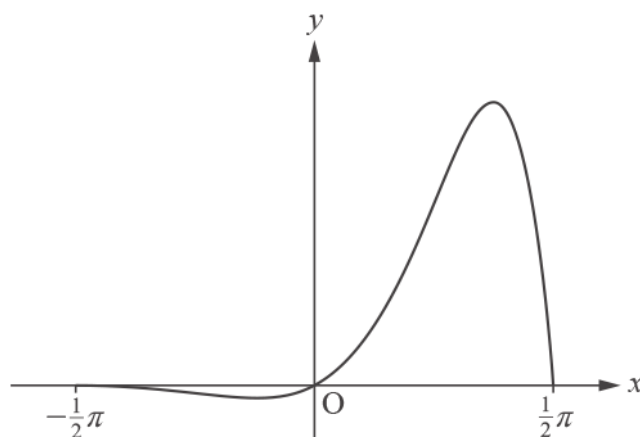


Fig. 13

Given that the maximum value of $f(x)$ occurs when $x = \frac{3}{8}\pi$, find the exact minimum value of $f(x)$.

[8]

- 24 Fig. 5 shows triangle ABC where $AB = 7\text{ cm}$, $BC = 8\text{ cm}$ and $AC = 5\text{ cm}$. The curve is an arc of a circle with centre C and radius $r\text{ cm}$.

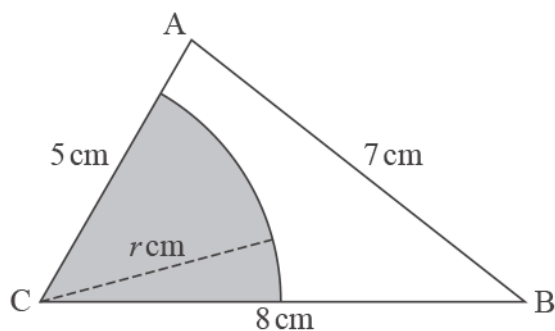


Fig. 5

Exactly half the area of the triangle is shaded. Find the value of r .

[5]

25 Rebecca is looking for the root of the equation



$\sin 2x^2 - \cos 5x = 0$

that lies between 0.2 and 0.3.

She uses the standard small angle approximations for $\sin \theta$ and $\cos \theta$ to find an estimate.

(a) Show that Rebecca's method gives an estimate of 0.26 when rounded to 2 decimal places.

[3]

(b) Use a change of sign method to determine whether this value gives an estimate of the root which is correct to 2 decimal places.

[2]

Rebecca then uses the Newton-Raphson method to find another estimate for the root.

(c) Show that using $x_0 = 0.2$ gives the value $x_1 = 0.291\,989\,3$ correct to 7 decimal places.

[5]

(d) Continue this method, showing the result of each iteration, to find the root correct to 3 significant figures.

[3]

(a) Prove that $(\operatorname{cosec} \theta - \cot \theta)^2 = \frac{1 - \cos \theta}{1 + \cos \theta}$.

[3]

(b) Hence solve the equation $(\operatorname{cosec} \theta - \cot \theta)^2 = \frac{1}{3}$ for $0^\circ < \theta < 360^\circ$.

[3]



- (a) Express $\sqrt{2} \cos x - \sin x$ in the form $R \cos(x + \alpha)$, where $0 < \alpha < \frac{\pi}{2}$.

[3]

- (b) You are given that

$$f(x) = \frac{5}{2 + \sqrt{2} \cos x - \sin x} \text{ for } 0 \leq x \leq 2\pi.$$

Find the minimum value of $f(x)$, giving your answer in the form $a + b\sqrt{c}$ where a , b and c are integers to be determined.

[3]

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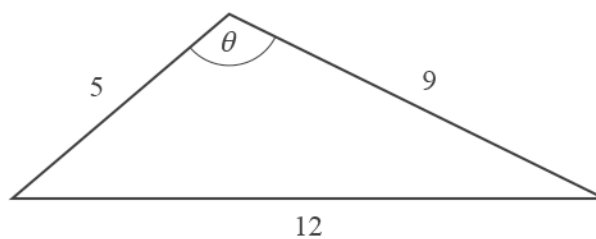


Fig. 6

Calculate the size of angle θ , giving your answer correct to the nearest degree.

[3]

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$$x = \tan \theta, y = 1 + \cos 2\theta, \text{ for } 0 \leq \theta \leq \pi \text{ with } \theta \neq \frac{1}{2}\pi.$$

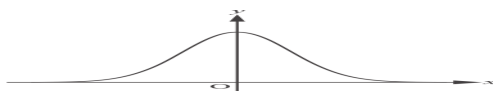


Fig. 9

- (a) Explain why $\theta \neq \frac{1}{2}\pi$ [1]

- (b) Show that the maximum value of y for points on the curve is 2. [1]

- (c) Show that the cartesian equation of the curve is $y = \frac{2}{1+x^2}$. [3]

- (d) In this question you must show detailed reasoning.

The point P in the first quadrant lies on the curve. Find the coordinates of P given that OP is the normal to the curve at P. [7]

- 30 Fig. 6 shows a circle with centre at the origin passing through the point A with coordinates (2, 0). The point B in the first quadrant lies on this circle.

The area of sector AOB is $\frac{2}{3}\pi$.

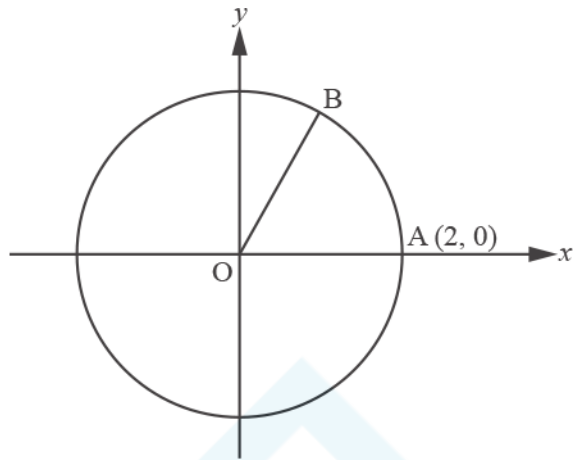


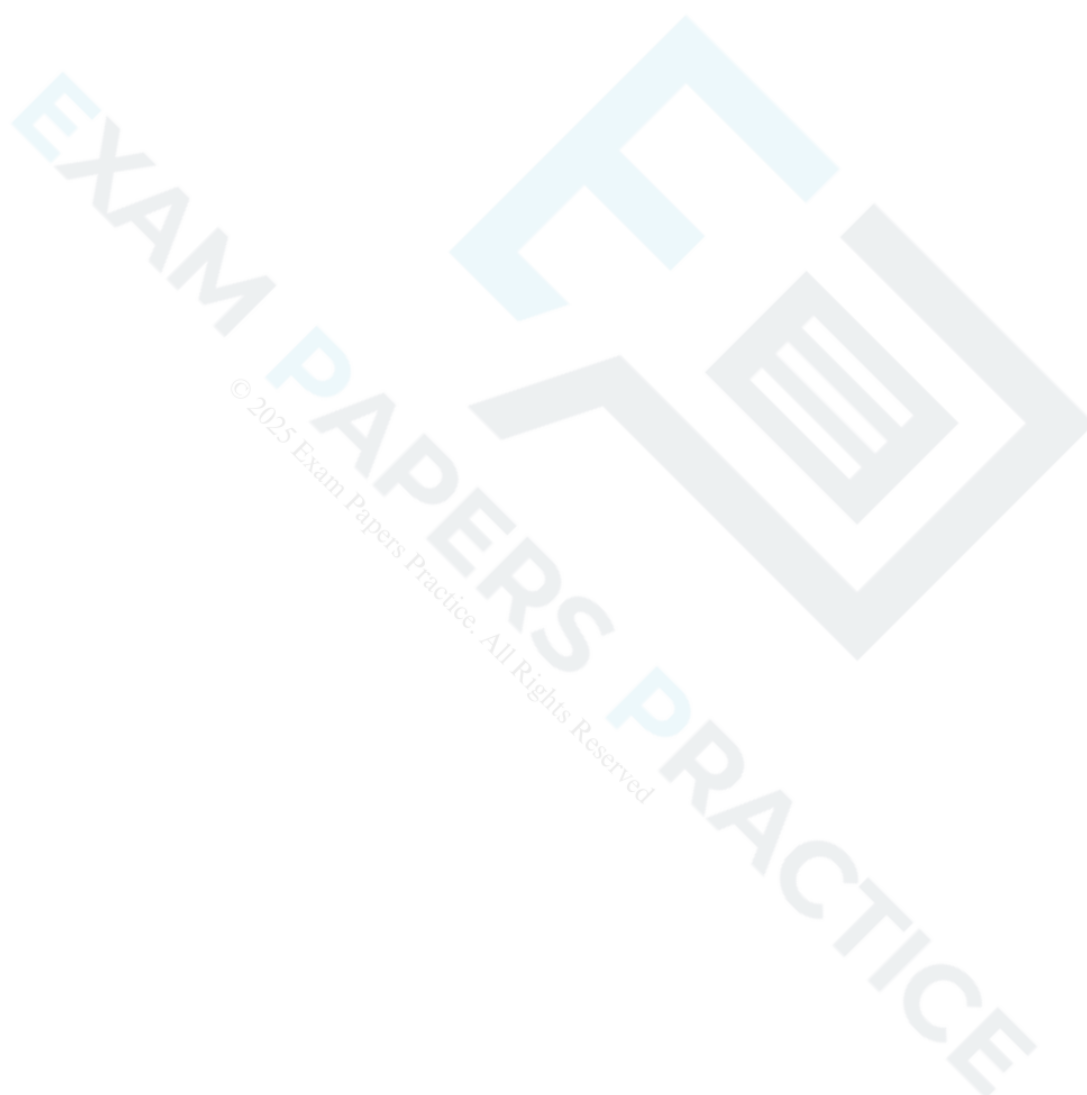
Fig. 6

- (a) Find the exact coordinates of B.

[4]

(b) The circle in Fig. 6 is reflected in line AB. Find the equation of the image circle.

[3]



31 (a) Sketch the graphs of $y = 4 \cos x$ and $y = 2 \sin x$ for $0^\circ \leq x \leq 180^\circ$ on the same axes.

[2]

(b) Find the exact coordinates of the point of intersection of these graphs, giving your answer in the form $(\arctan a, k\sqrt{b})$, where a and b are integers and k is rational.

[4]

(c) A student argues that without the condition $0^\circ \leq x \leq 180^\circ$ all the points of intersection of the graphs would occur at intervals of 360° because both $\sin x$ and $\cos x$ are periodic functions with this period. Comment on the validity of the student's argument.

[1]

32 (See Insert for Jun18 64003.)



(a) In Fig. C1.3, angle $CBD = \theta$. Show that angle CDA is also θ , as given in line 23.

[2]

(b) Prove that $h = \sqrt{ab}$, as given in line 24.

[2]

- 33 Fig. 3 shows a circle with centre O and radius 1 unit. Points A and B lie on the circle with angle $AOB = \theta$ radians. C lies on AO, and BC is perpendicular to AO.

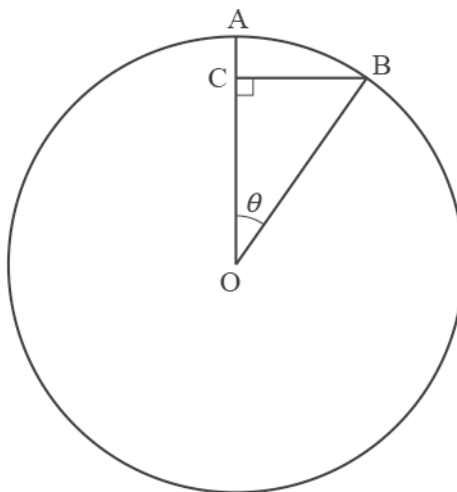


Fig. 3

Show that, when θ is small, $AC \approx \frac{1}{2}\theta^2$.

[2]

34 Triangle ABC is shown in Fig. 1.

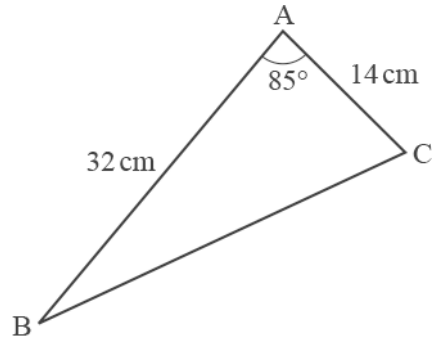


Fig. 1

Find the perimeter of triangle ABC.

[3]



- (a) Express $7\cos x - 24\sin x$ in the form $R \cos(x + \alpha)$, where $0 < \alpha < \frac{\pi}{2}$.

[3]

- (b) Write down the range of the function

$$f(x) = 12 + 7\cos x - 24\sin x, \quad 0 \leq x \leq 2\pi.$$

[2]

- 36 Fig. 12 shows the circle $(x - 1)^2 + (y + 1)^2 = 25$, the line $4y = 3x - 32$ and the tangent to the circle at the point A (5, 2). D is the point of intersection of the line $4y = 3x - 32$ and the tangent at A.

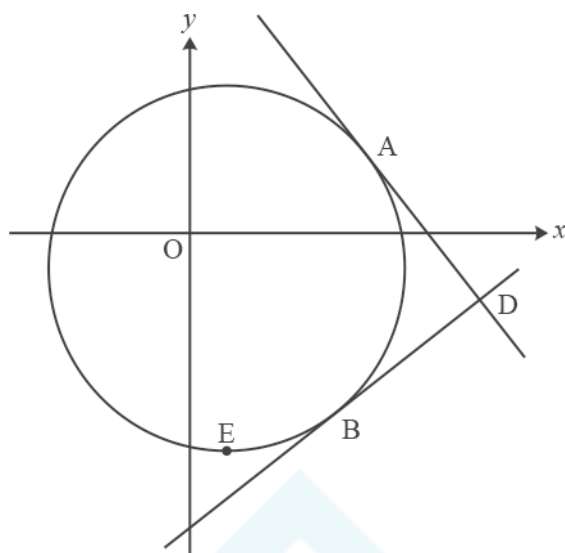


Fig. 12

- (a) Write down the coordinates of C, the centre of the circle.

[1]

- (b) (i) Show that the line $4y = 3x - 32$ is a tangent to the circle.

[4]



(ii) Find the coordinates of B, the point where the line $4y = 3x - 32$ touches the circle.

[1]

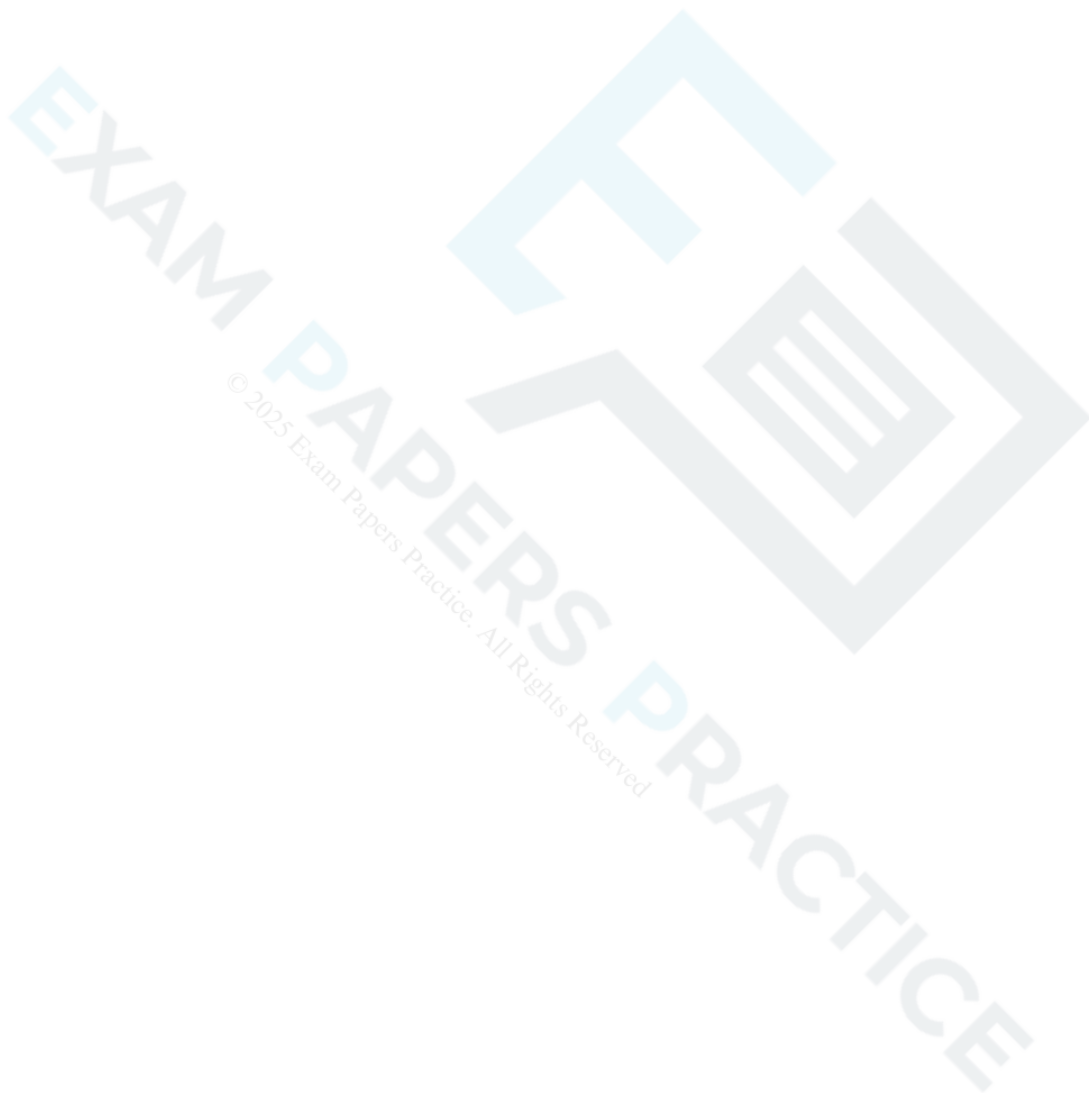
(c) Prove that ADBC is a square.

[3]

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(d) The point E is the lowest point on the circle. Find the area of the sector ECB.

[5]



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37 (a) Show that $8 \sin^2 x \cos^2 x$ can be written as $1 - \cos 4x$.

[3]

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(b) Hence find $\int \sin^2 x \cos^2 x \, dx$.

[3]

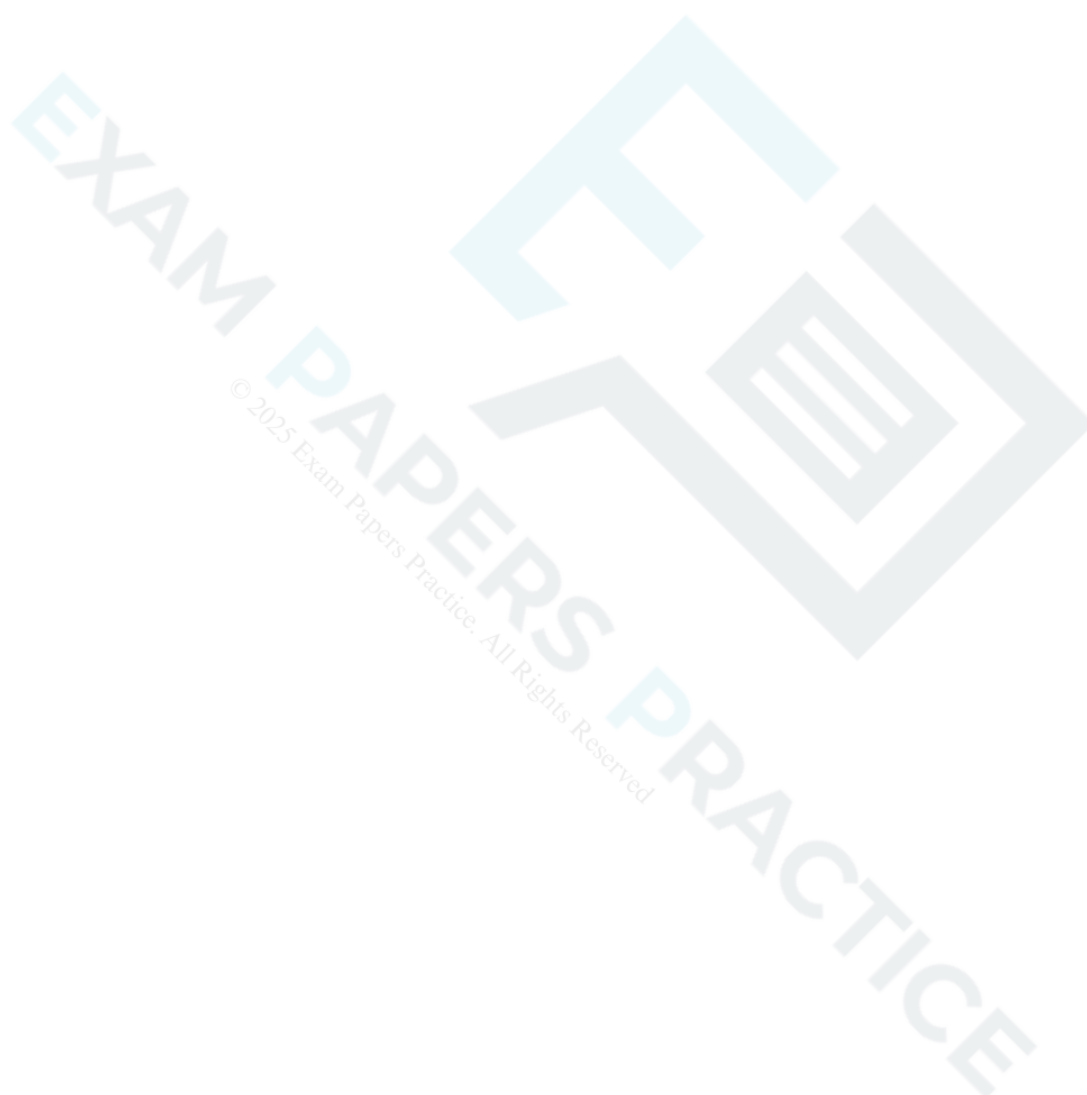
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38 In this question you must show detailed reasoning.

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Solve the equation $\sec^2 \theta + 2 \tan \theta = 4$ for $0^\circ \leq \theta < 360^\circ$.

[4]



- 39 In an experiment 500 fruit flies were released into a controlled environment. After 10 days there were 650 fruit flies present.

Munirah believes that N , the number of fruit flies present at time t days after the fruit flies are released, will increase at the rate of 4.4% per day. She proposes that the situation is modelled by the formula $N = Ak^t$.

(a) Write down the values of A and k .

[2]

(b) Determine whether the model is consistent with the value of N at $t = 10$.

[2]

(c) What does the model suggest about the number of fruit flies in the long run?

[1]

Subsequently it is found that for large values of t the number of fruit flies in the controlled environment oscillates about 750. It is also found that as t increases the oscillations decrease in magnitude.

Munirah proposes a second model in the light of this new information.

$$N = 750 - 250 \times e^{-0.092t}.$$

(d) Identify three ways in which this second model is consistent with the known data.

[3]

(e) (i) Identify one feature which is not accounted for by the second model.

[1]

(ii) Give an example of a mathematical function which needs to be incorporated in the model to account for this feature.

[1]

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40 In this question you must show detailed reasoning.

Fig. 8 shows the curve with parametric equations

$$x = 7 \cos \theta + 2 \cos 2\theta, y = 2 + \sin \theta, (0 \leq \theta \leq 2\pi).$$

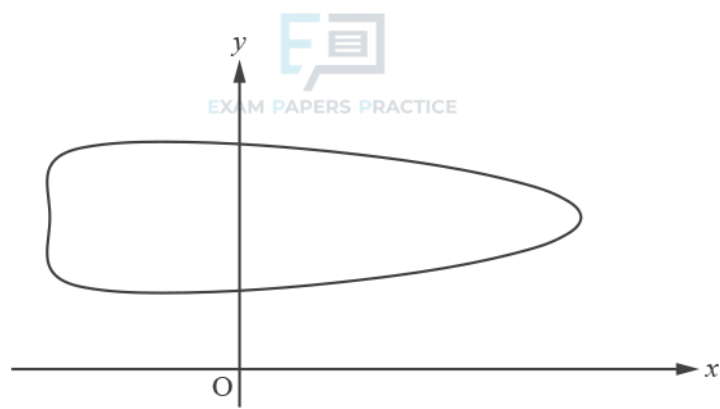


Fig. 8

(a) Find the coordinates of the point on the curve with the greatest y -coordinate.

[4]

(b) Determine the exact y -coordinates of the points where the curve crosses the y -axis.

[6]

- 41 Fig. 7 shows a circle with centre A and radius 6 cm, and a circle with centre B and radius 2 cm. The two circles touch. DCE is a common tangent to the circles. ABE is a straight line.

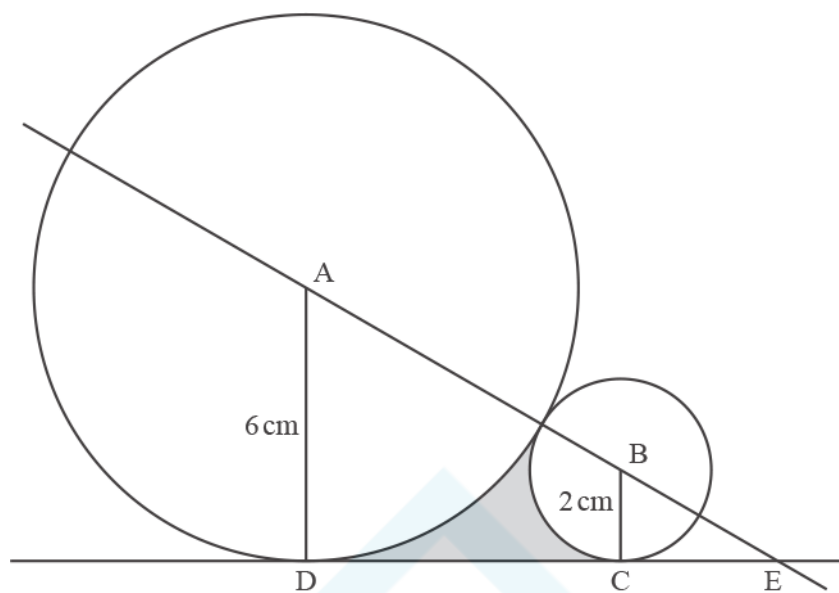


Fig. 7

- (a) Give a reason why angles ADE and BCE are right angles.

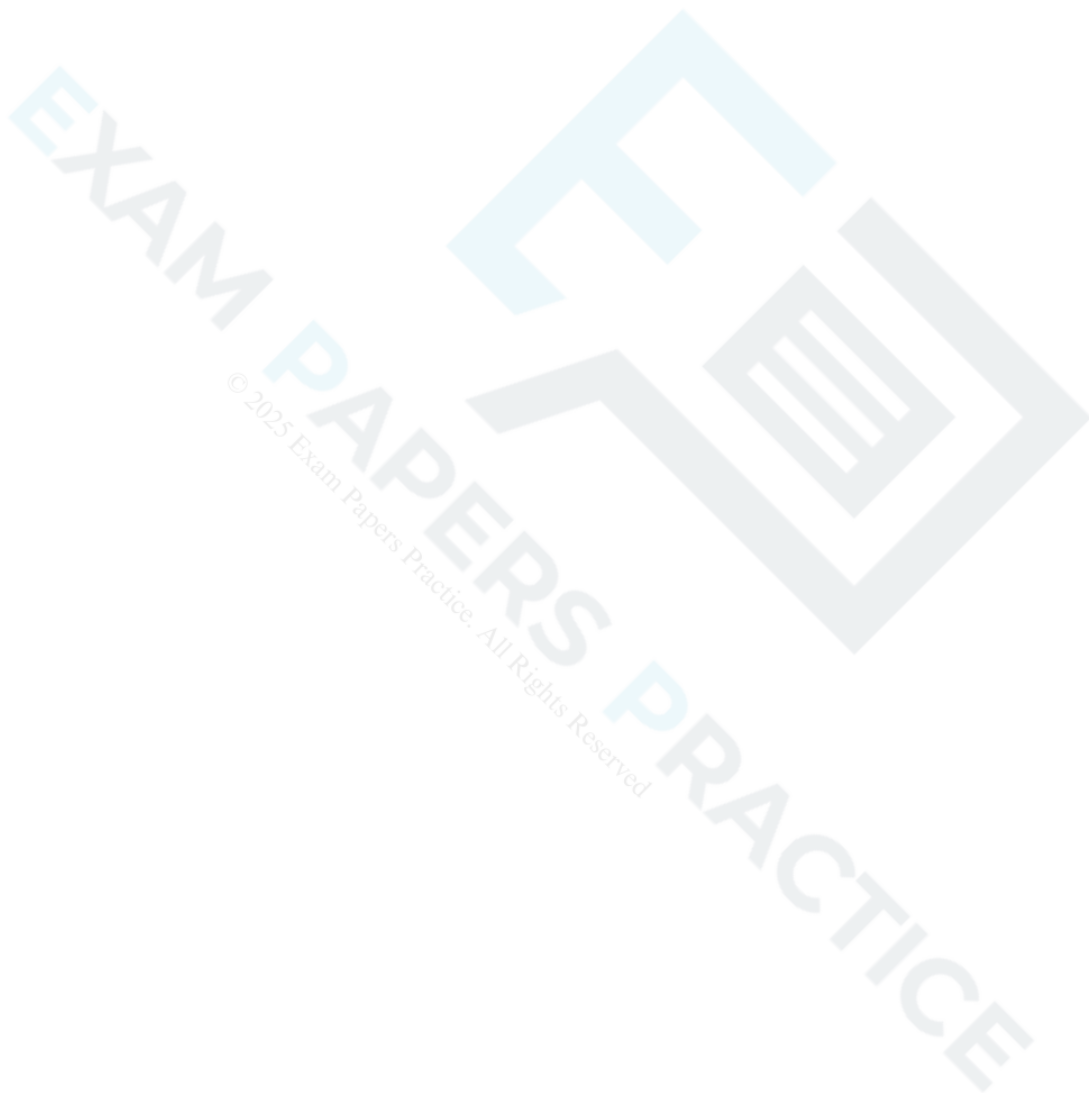
[1]

- (b) Show that angle DAB is $\frac{1}{3}\pi$ radians.

[2]

(c) Show that trapezium ABCD has area $16\sqrt{3} \text{ cm}^2$.

[4]



(d) Calculate the area of the shaded region enclosed by the two circles and the common tangent DCE. Give your answer correct to 3 decimal places.

[3]

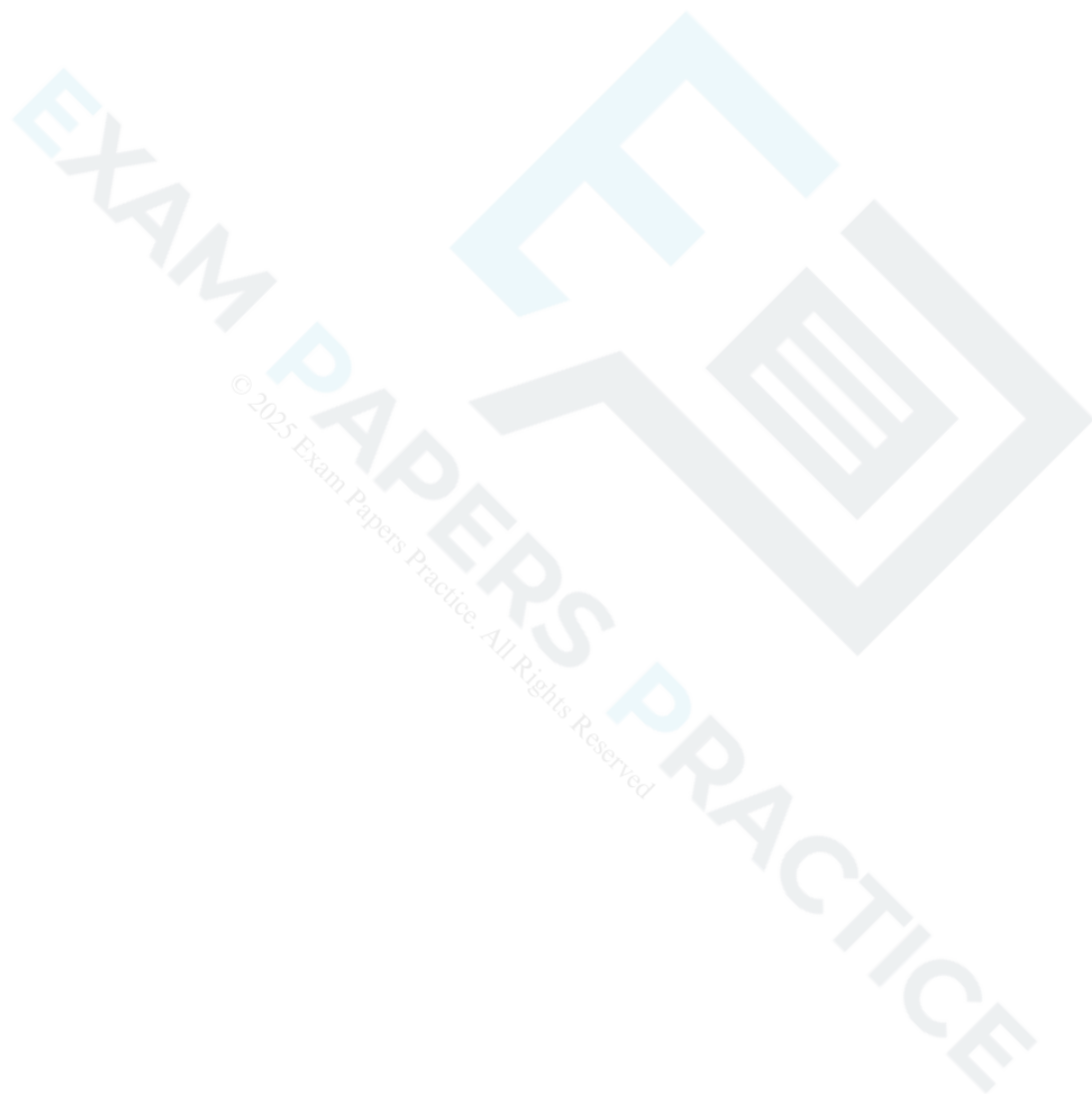


Fig. 6 shows the curves $y = \sin x$ and $y = \cos x$ for $-\pi \leq x \leq \pi$.

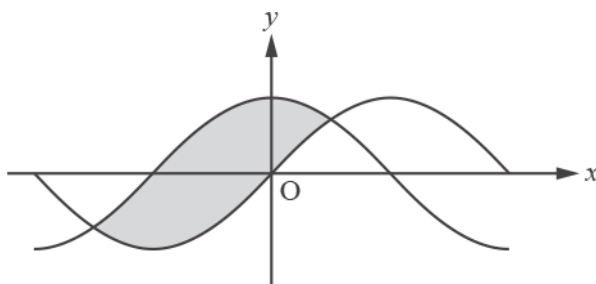
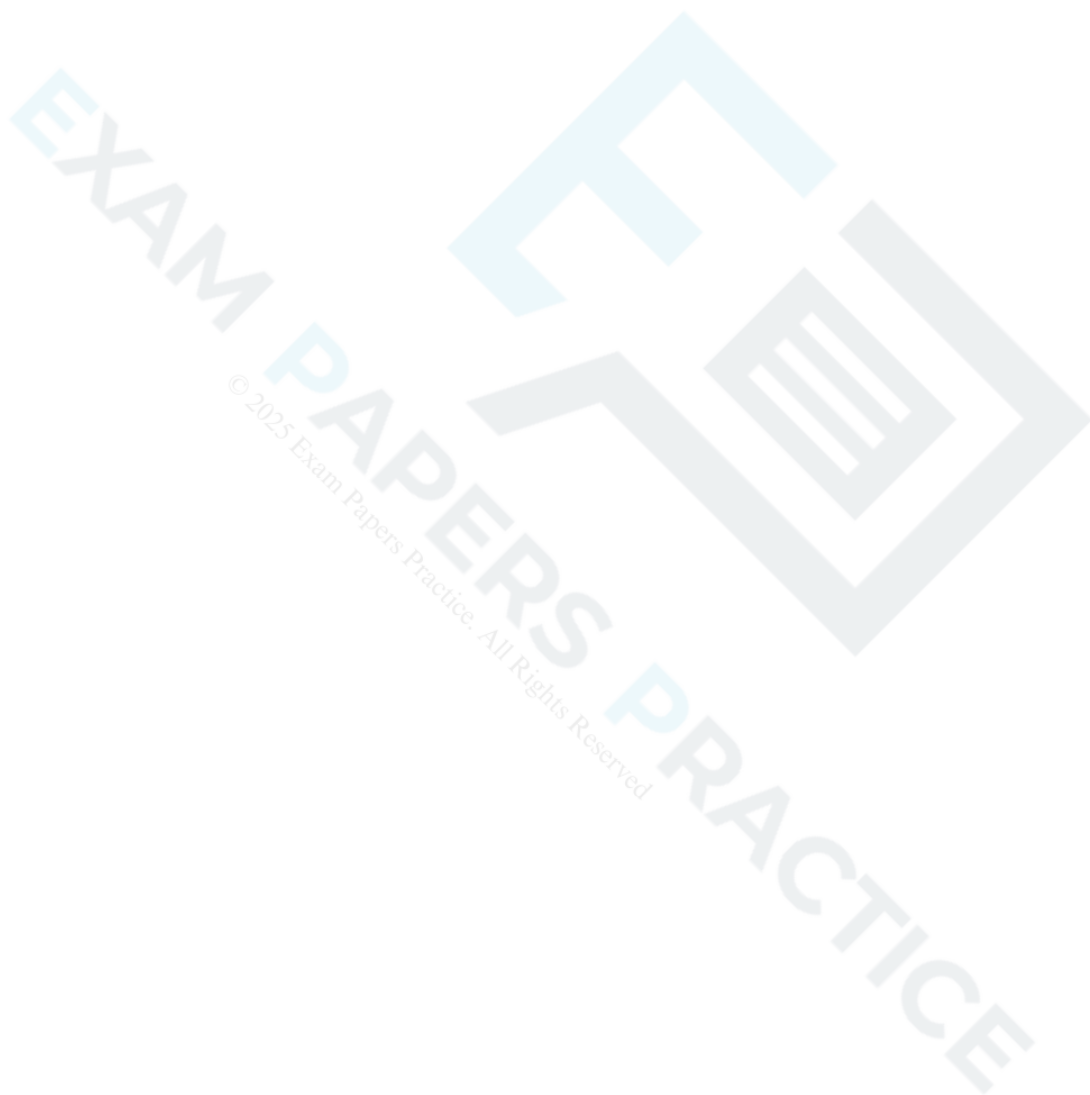


Fig. 6

- (a) Show that the area of the region enclosed between the curves is given by $\int_a^b (\cos x - \sin x) dx$, where the values of a and b are to be determined. [4]

(b) Find the exact value of the area of this region.

[3]



(a) Express $8 \cos x + 5 \sin x$ in the form $R \cos(x - \alpha)$, where R and α are constants with $R > 0$ and

[3]

$$0 < \alpha < \frac{1}{2}\pi.$$

(b) Hence solve the equation $8 \cos x + 5 \sin x = 6$ for $0 \leq x < 2\pi$, giving your answers correct to 4 decimal places.

[3]

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44 Fig. 7 shows an isosceles triangle ABC, in which $AC = BC = a$ and angle $ACB = \theta$.

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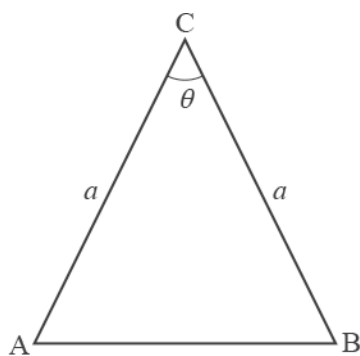


Fig. 7

(a) Use the cosine rule to write down an expression for AB^2 in terms of a and θ .

[1]

(b) D is the midpoint of AB. Use an expression for AD in terms of a and θ to show that

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$$\cos \theta \equiv 1 - 2 \sin^2\left(\frac{1}{2}\theta\right).$$

[3]

The identity in part (b) is valid for all values of θ . However, the argument in parts (a) and (b) does not prove the identity for all values of θ .

(c) State the values of θ for which the argument is valid.

[1]

- 46 A field is to be turned into a car park, a pond and a meadow. Fig. 10 shows one possible design.

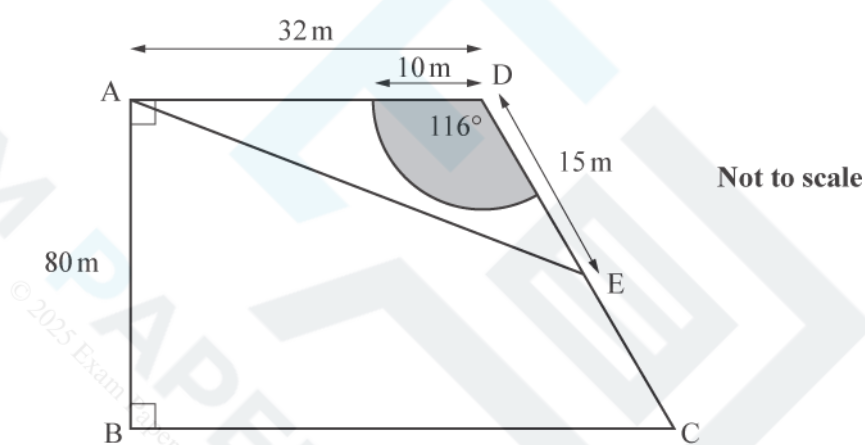


Fig. 10

The field ABCD is a trapezium, with sides AD and BC parallel. AD = 32 m, AB = 80 m, angle B = 90° and angle D = 116° . The pond, shown shaded, is a sector of a circle, centre D and radius 10 m. The point E is on DC, with DE = 15 m.

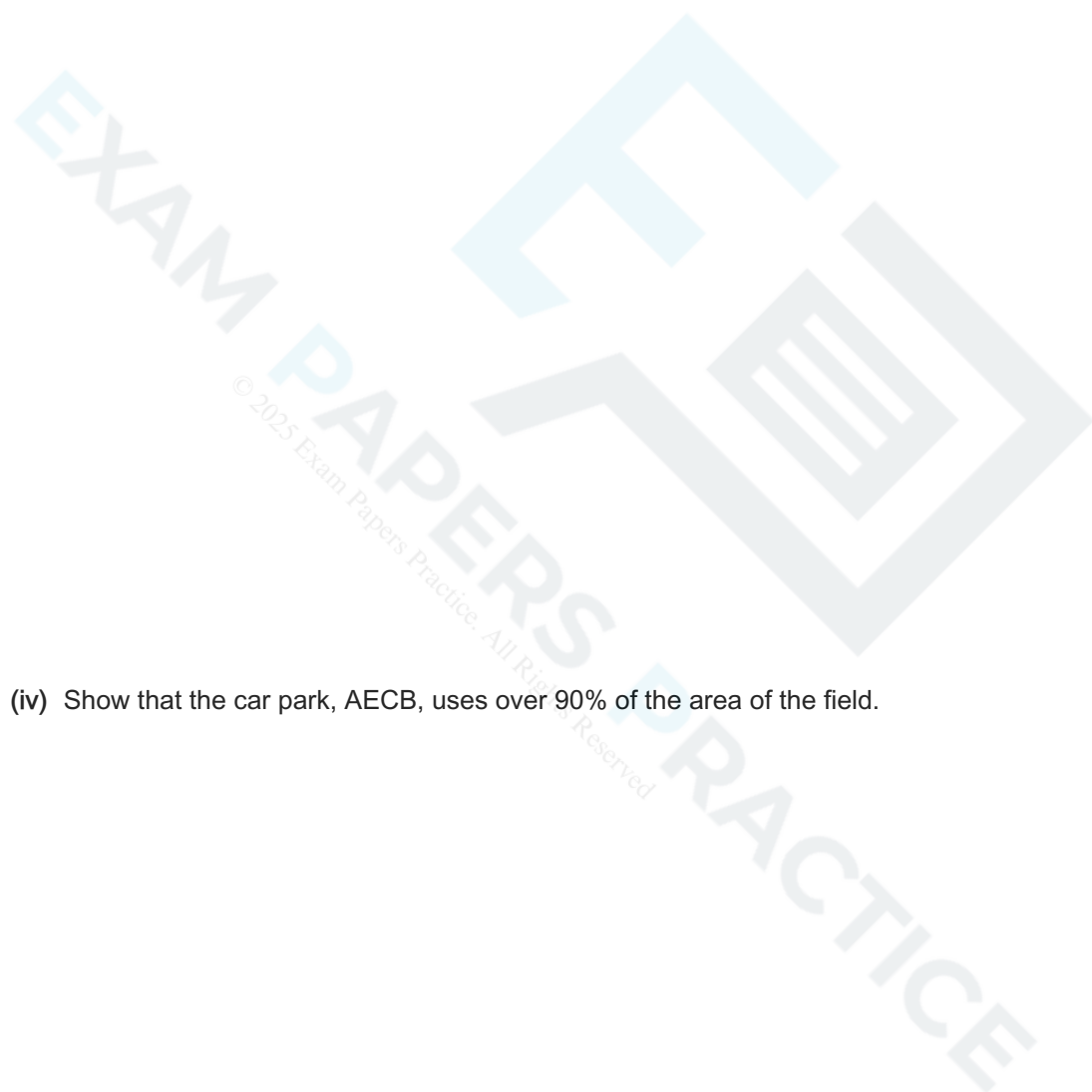
- (i) Calculate the length of AE.

[2]

- (ii) Calculate the perpendicular distance of AE from D. Hence verify that the pond lies entirely within triangle ADE. [3]

The meadow is the triangle ADE except for the pond.

- (iii) Calculate the area of the pond and the area of the meadow [4]



(iv) Show that the car park, AECB, uses over 90% of the area of the field.

[4]

- 47 You are given that $6 \cos^2 x = 5 - \sin x$, where x is in radians. Show that $6 \sin^2 x - \sin x - 1 = 0$. Solve this equation for $0 \leq x \leq 2\pi$.

[5]



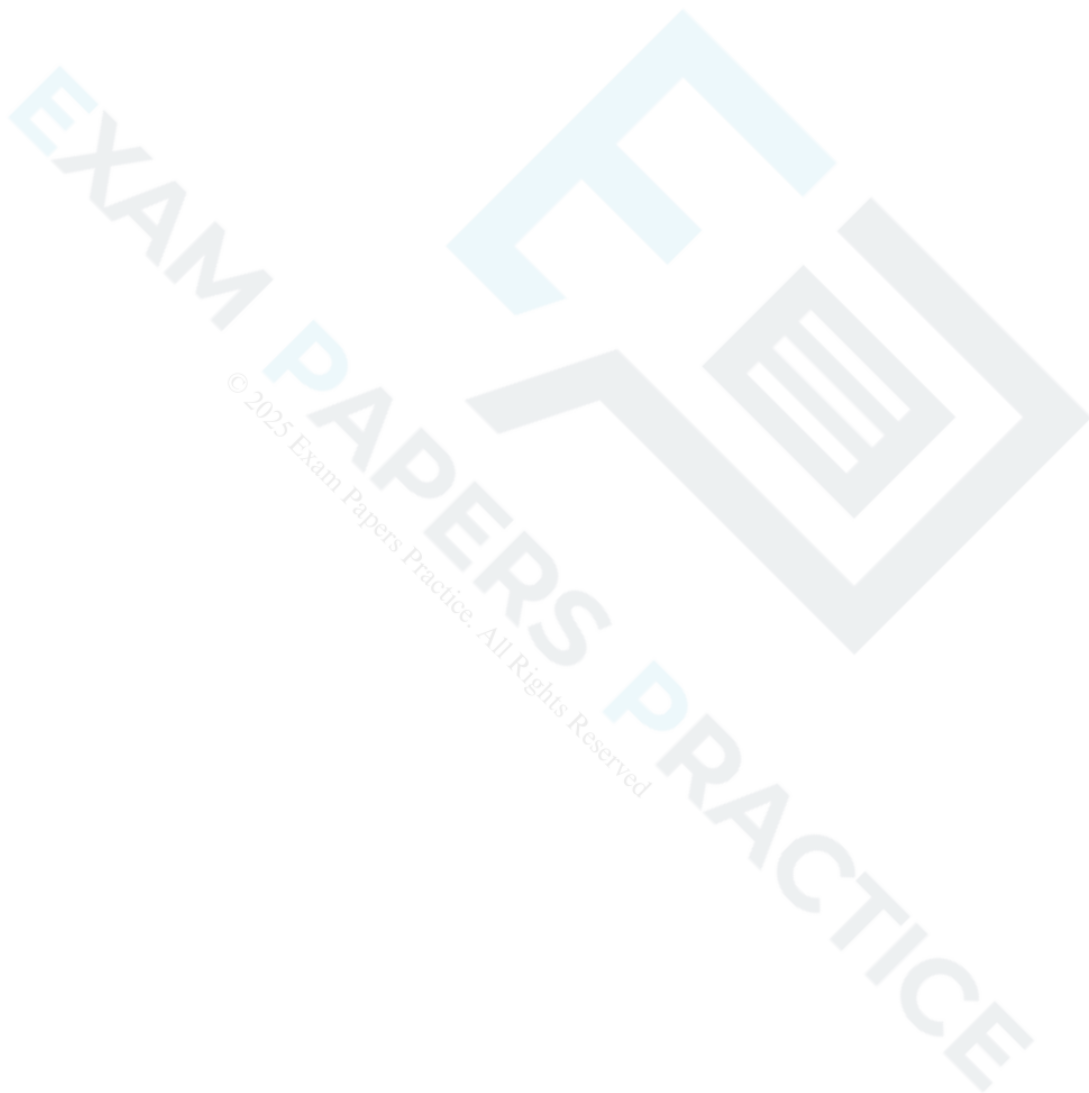
48 In this question you must show detailed reasoning.

A curve has parametric equations

$$x = \cos t - 3t \text{ and } y = 3t - 4 \cos t - \sin 2t, \text{ for } 0 \leq t \leq \pi.$$

Show that the gradient of the curve is always negative.

[7]



49 The sides of a triangle are of length 47, 53 and 94 units. Calculate the size of the largest angle.

[3]

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50 (a) Write down the exact values of $\tan 45^\circ$ and $\tan 60^\circ$.

[1]

(b) In this question you must show detailed reasoning.

Show that $\tan 15^\circ = 2 - \sqrt{3}$.

[4]

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- (a) Express $2 \cos \theta + 3 \sin \theta$ in the form $R \sin(\theta + \alpha)$, where $0 < \alpha < \frac{1}{2}\pi$ and R is a positive constant given in exact form. [4]

- (b) Determine the set of values of k for which the curve $y = k + 2 \cos x + 3 \sin x$ lies completely above the x -axis.

[4]

- (c) Explain why the curve $y = \frac{1}{k + 2 \cos x + 3 \sin x}$ lies completely above the x -axis for the set of values of k found in part (b).

[1]

- 52 Gerry has been collecting data about the ladybird population in his locality, where they appear mainly during the spring, summer and autumn. He has carried out surveys on many days over a three-month period in the summer, and has recorded the values of N , the number of ladybirds seen, and t , the time in days after 1 January. Using graph-drawing software, he has found that the part of the quadratic curve shown in Fig. 13.1 is a good fit for his data.

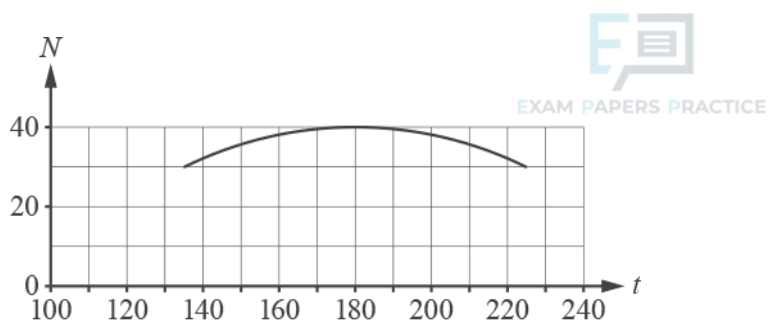


Fig. 13.1

This curve has a maximum point at (180, 40) and the point (135, 30) also lies on the curve. Gerry uses the equation of the curve in the form

$$N = a - b(t - c)^2$$

as a model for the ladybird population.

- (a) Find the values of a and c and show that $b = \frac{2}{405}$.

[3]

- (b) (i) Find the number of ladybirds predicted by the model when $t = 250$.

[1]

- (ii) Explain why this predicted value may be unreliable.

[1]

- (c) (i) Explain why the model is not valid for $t < 90$.

[1]

(ii) Give another range of values of t for which the model is not valid.

[1]

With the software set to degrees, Gerry also produces the graph in Fig. 13.2 to model his data for suitable values of t . This graph is a transformation of $N = \cos t$; it passes through $(90, 0)$ and has a maximum point at $(180, 40)$.

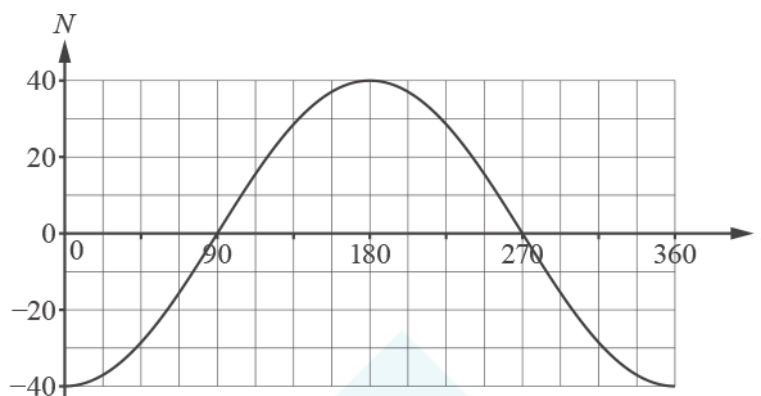


Fig. 13.2

(d) Describe a sequence of transformations that maps the curve $N = \cos t$ to the curve in Fig. 13.2.

[2]

(e) Write down the equation for N in terms of t for this model.

[1]

(f) (i) Explain how the model can be refined to have a period of 365 days.

[1]

(ii) What effect, if any, does this refinement have on the maximum point of the graph?

[1]

53 (a) Prove the identity $\sec\theta - \cos\theta \equiv \tan\theta \sin\theta$.



EXAM PAPERS PRACTICE

[4]

(b) Hence or otherwise solve the equation $\sec\theta - \cos\theta = \frac{1}{2}\tan\theta$ for $0 \leq \theta < 2\pi$.

[4]

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- 54 A globe hangs on the end of a light wire that makes an angle of θ with the vertical. It is held in place by a light horizontal string, as shown in Fig. 7.

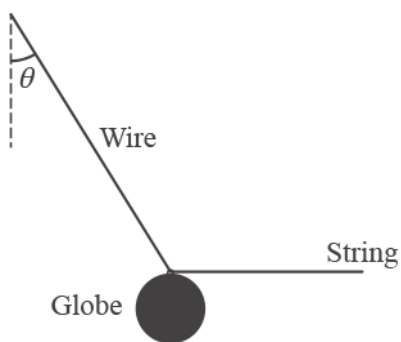


Fig. 7

Find the set of values of θ for which the tension in the string is less than the weight of the globe.

[4]

- 55 Fig. 6.1 shows a shape consisting of an isosceles triangle ABC and a semicircle with AC as diameter. Angle $ABC = \frac{1}{6}\pi$ radians and $AB = BC = 10$ cm.

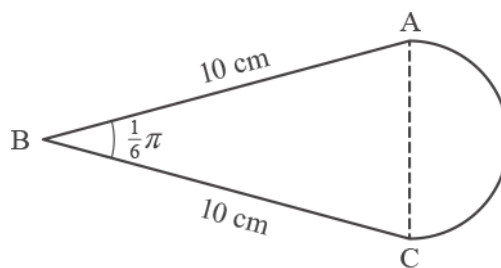


Fig. 6.1

- (a) Calculate the total area of the shape.

[4]

Fig. 6.2 shows a sector DEF of a circle of radius 10 cm. Angle DEF = θ radians.

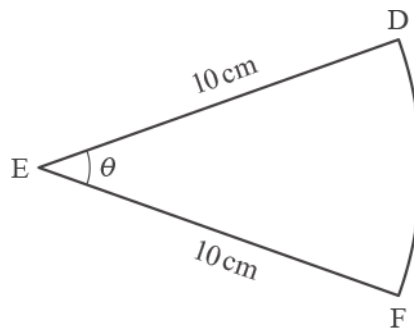


Fig. 6.2

- (b) The shapes in Fig. 6.1 and Fig. 6.2 have the same area. Find θ , giving your answer correct to 3 significant figures.

[2]

- 56 A car travels along a straight track for 5 seconds. Its displacement s metres after t seconds is given by

$$s = 3t + 0.1t^3.$$

Show that the car does not have constant acceleration.

[3]

END OF QUESTION PAPER



Question			Answer/Indicative content	Marks	Guidance
1		i	$x = 0$ $x = \pi$	B1 (AO1.1) B1 (AO1.1) [2]	If answers given in both degrees and radians follow inst 2g 180 gets 0 <u>Examiner's Comments</u> In Question 5 many did not realise the need to use radians. The command word used in part (d) is 'Determine' so, when considering whether the iteration converges to 'this root', the answer should say either 'Therefore it does converge to this root' or 'No, it actually converges to ...'.
		ii	$(\frac{\pi}{2}, 1)$	B1 (AO2.2a) [1]	(90, 1) or (1.57, 1) get 0
			Total	3	

Question			Answer/Indicative content	Marks	Guidance	
2			$\frac{1}{2} - \sin 2x \cos x = \sin x \cos 2x$ $\sin 3x = \frac{1}{2}$ oe $x = \frac{\pi}{18}$ and $x = \frac{5\pi}{18}$ $\pm \int \left(\sin x \cos 2x - \left(\frac{1}{2} - \sin 2x \cos x \right) \right) dx$ oe $F[x] = -\frac{x}{2} - \frac{\cos 3x}{3}$ $F\left[\frac{5\pi}{18}\right] - F\left[\frac{\pi}{18}\right]$ $\frac{\sqrt{3}}{3} - \frac{\pi}{9}$ or $\frac{3\sqrt{3}-\pi}{9}$ cao	M1 (AO3.1a) M1 (AO2.1) A1 (AO3.2a) A1 (AO1.1) M1 (AO1.1) A1 (AO1.1) M1 (AO1.1) A1 (AO3.2a) [8]	from compound angle formula allow sign errors only A1 for each ignore limits allow the positive of this F[x] must be one of the correct forms	or $4\sin^3 x - 3\sin x + \frac{1}{2} = 0$ oe 0.17453... A1 0.87266... A1 to 2 or more sf $\pm \left(-\frac{4}{3}\cos^3 x + \cos x - \frac{x}{2} \right)$ oe or $\pm \left(-\frac{1}{3}\cos 2x \cos x + \frac{1}{3}\sin x \sin 2x - \frac{1}{2}x \right)$ oe for A1 F[0.87266] – F[0.17453] for M1
Total				8	<u>Examiner's Comments</u> Candidates who did well either spotted the compound angle application or were able to derive the correct cubic in sin x when finding the points of intersection of the curve. They recognised how to apply integration and some were able to do this correctly. Those who did less well were unable to make progress with the integration or slipped up in forming the cubic in sin x. Candidates who did not do well were unable to progress from setting the two equations equal or from stating the correct integration needed.	

Examiner's Comments

Candidates who did well either spotted the compound angle application or were able to derive the correct cubic in $\sin x$ when finding the points of intersection of the curve. They recognised how to apply integration and some were able to do this correctly. Those who did less well were unable to make progress with the integration or slipped up in forming the cubic in $\sin x$. Candidates who did not do well were unable to progress from setting the two equations equal or from stating the correct integration needed.

Question		Answer/Indicative content	Marks	Guidance
3	a	$\sin 2x \approx 2x$ or $\sin x \approx x$ used $\int \left(\frac{1}{x}\right) dx$ or $\int \left(\frac{1}{x} - x\right) dx$ obtained oe nfww $F[x] = \ln x$ oe or $F[x] = \ln x - \frac{1}{2} x^2$ oe $\ln(0.05) - \ln(0.01) = \ln 5$ oe or $\ln(0.05) - \ln(0.01) + 0.0012 \approx \ln 5$ oe	M1 (AO3.1a) A1 (AO1.1) A1 (AO1.1) A1 (AO3.2a) [4]	may see $\cos x \approx 1 - \frac{x^2}{2}$ intermediate step needed from here to earn final mark
	b	differentiation of <i>their</i> $\frac{1}{x}$ substitution of 0.01 and $-10\,000$ correctly obtained	M1 (AO2.1) A1 (AO1.1) [2]	or differentiation of y using quotient rule and use of small angle approximation from $-\frac{1}{x^2}$ or $-\frac{1}{x^2} - 1$ oe

Question			Answer/Indicative content	Marks	Guidance
	c		4.54066×10^{-6} or 0.00000454066 cao (no sign change for 6 dp), but sign change for 5 dp or last two iterates agree to 5dp 0.94775	B1 (AO2.5) E1 (AO3.1a) B1 (AO3.2a) [3]	allow sign change between 0.947745 and 0.9477475 <u>Examiner's Comments</u> Candidates who did well in this question made efficient use of the small angle approximation for $\sin x$ and sensibly used $\sin 2x \approx 2x$ to obtain $\int \frac{1}{x} dx$, or used the double angle formula and then the small angle approximation for x . Full marks in parts (a) and (b) was commonly achieved. Those who did less well did not simplify their expression for $\frac{1}{x}$ fully before integrating and went on to make a slip (often integrating by parts) and similarly used the product or quotient rule in part (b) to differentiate $\frac{x}{x^2}$ for example. Candidates who did well in part (c) were clearly familiar with spreadsheets and were able to give a clear and correct response to both requests. Candidates who did less well either didn't understand the spreadsheet notation, or didn't appreciate the significance of the change of sign in the table.
			Total	9	

Question			Answer/Indicative content	Marks	Guidance
4	a		$4 + 8\cos 8x$	M1* (AO1.1) A1 (AO1.1) [2]	differentiation with either term correct all correct
	b		attempt to solve their $4 + 8\cos 8x = 0$ $\frac{\pi}{12}$ isw cao	M1dep* (AO1.1) A1 (AO1.1) [2]	one intermediate step seen Examiner's Comments Candidates who did well in Question 3 differentiated successfully in part (a) and set their derivative equal to 0 in part (b). They understood that the answer should be presented in radians. Those who did less well made a slip in part (a)– usually omitting an 8 – or worked in degrees in part (b). The very few candidates who did not do well were unable to differentiate either term successfully in part (a).
			Total	4	
5	a		$\frac{1}{2} \times 8^2 \times 2.1$ 67.2 or 67 cao	M1 (AO1.1a) A1 (AO1.1) [2]	$\theta = 120 - 120.3211\dots$ or $\frac{\theta^\circ}{360} \times \pi \times 8^2$ ignore units
	b		16.8 or 17 cao	B1 (AO1.1) [1]	ignore units
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
6			$\frac{1}{2} \times 22.1 \times 18.0 \times \sin 133^\circ$ 14 5 cao	M1 (AO1.1a) A1 (AO1.1) [2]	or $\frac{1}{2} \times 18.0 \times 36.8 \times \sin 26^\circ$ or $\frac{1}{2} \times 22.1 \times 36.8 \times \sin 21^\circ$ ignore units NB 172.8 or 173 unsupported implies M1 Examiner's Comments Question 1 was very well done by most candidates. Those who did less well incorrectly rounded to 146 or presented their final answer to a different precision to the request for 3 sf. A few candidates appeared to have used their calculator in radian mode and did not correctly find $\sin(133^\circ)$.
			Total	2	

Question			Answer/Indicative content	Marks	Guidance	
7			$3 \cos \theta + 8 \frac{\sin \theta}{\cos \theta} [= 0]$ $3 \cos^2 \theta + 8 \sin \theta = 0$ $3(1 - \sin^2 \theta) + 8 \sin \theta = 0$ $3 \sin^2 \theta - 8 \sin \theta - 3 = 0$ $\sin \theta = -\frac{1}{3}$ WWW $\theta = 341, 199$	M1* (AO1.2) M1dep* (AO1.1) M1 (AO2.4) A1 (AO1.1) A1 (AO1.1a) A1 (AO2.4) [6]	multiplication of all terms of their equation by $\cos \theta$ dependent on award of previous M1 from factorising or quadratic formula No wrong values; accept 199.47, 340.53	Condone sign errors and number errors for the three M marks May see $\sin \theta = 3$
			Total	6		

Examiner's Comments

Almost all candidates were able to begin this question by replacing $\tan \theta$ with

$\frac{\sin \theta}{\cos \theta}$, and then going on to multiply

through the equation by $\cos \theta$. Some candidates lost the '8' or arrived at '24', and only a minority ended up with the correct quadratic equation in $\sin \theta$. Most arriving at this point were able to discard the root of 3, though not all could correctly derive the 2 correct angles from

$$\sin^{-1}\left(-\frac{1}{3}\right).$$

Question	Answer/Indicative content	Marks	Guidance
8	$\frac{dx}{d\theta} = \sec^2 \theta$ $\int_{x=-1}^{x=1} \frac{\sec^2 \theta}{(1 + \tan^2 \theta)^2} d\theta$ $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sec^2 \theta}{(1 + \tan^2 \theta)^2} d\theta$ $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{\sec^2 \theta} d\theta$ $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos^2 \theta d\theta$ $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{2} (\cos 2\theta + 1) d\theta$ $\frac{1}{2} \left[\frac{1}{2} \sin 2\theta + \theta \right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}}$ $\frac{1}{2} \left(1 + \frac{\pi}{2} \right) = \frac{1}{2} + \frac{\pi}{4}$	M1 (AO1.1a) M1 (AO1.1) M1 (AO1.1) M1 (AO2.2a) M1 (AO1.1) M1 (AO3.1a) M1 (AO1.1) A1 (AO2.1) [8]	o.e. Condone $\sec^2 x$ Substitution for either x or dx (limits may be missing) Limits (may be done at any point) Use of $1 + \tan^2 \theta = \sec^2 \theta$ Use of $\sec \theta = \frac{1}{\cos \theta}$ Use of $\cos 2\theta = 2 \cos^2 \theta - 1$ Integration. Must include at least one trig term. Limits may be wrong or missing Exact form
	Total	8	
9	Cosine rule in triangle ABC $AC^2 = 5^2 + 6^2 - 2 \times 5 \times 6 \cos 60^\circ = 31$ Sine rule in triangle ACD $\frac{AD}{\sin 60^\circ} = \frac{AC}{\sin 45^\circ}$ $AD = \frac{\sqrt{3}}{2} \times \sqrt{31} \times \sqrt{2} = \frac{1}{2} \sqrt{186}$	M1(AO1.1a) A1(AO1.1) M1(AO1.1a) A1(AO1.1) [4]	attempt to use the cosine rule Need not be evaluated soi Using their AC (not AC^2 in sine rule) oe. Must be surd form FT their AC
	Total	3	

Question			Answer/Indicative content	Marks	Guidance	
10	a		$h_{\max} = 5.15 + 3.4 \times 1 = 8.55$ $h_{\min} = 5.15 - 3.4 \times 1 = 1.75$ These are the correct h values for high and low tide	B1(AO3.4) [1]	Choosing $\cos t = \pm 1$ to give both values must be seen Allow without further comment	Allow for using given h values to find $\cos t = \pm 1$ only if there is a comment that these are max and min values for $\cos t$
	b	i	When $t = 1$ $8.55 = 5.15 + 3.4 \cos(a + b)$ So $\cos(a + b) = 1$ giving $a + b = 0$	B1(AO3.3) [1]	Correctly relating high tide, $t = 1$ and $\cos 0$	Accept 8.55 or $\cos t = 1$ as evidence of high tide
		ii	Minimum when $(at + b) = 180^\circ$ and $t = 7\frac{1}{3}$ So $\frac{22}{3}a + b = 180$	B1(AO3.3) [1]	Condone the use 7.2 hours here	Allow for $1.75 = 5.15 + 3.4 \cos\left(\frac{22}{3}a + b\right)$
		iii	Solve simultaneously to give $a = 28.42$ to 2 dp	M1(AO3.3) A1(AO3.3) [2]	Attempt to solve simultaneous equations: may be BC AG (value of b not needed here)	$[b = -28.42]$
	c		Substitute $h = 3$ $3 = 5.15 + 3.4 \cos(28.4t - 28.4)$ $\cos(28.4t - 28.4) = -\frac{43}{68}$ $28.4t - 28.4 = 129.2, \quad 230.8$ $t = 5.55, \quad 9.13$ He does not sail between 5.33 am and 9.08 am	M1(AO3.4) A1(AO3.4) A1(AO3.2a) [3]	Attempting to solve trig equation or inequality At least one correct [decimal] value for t Both times correct. Need not convert to hours and minutes. Must indicate between these times	

Question		Answer/Indicative content	Marks	Guidance	
	d	EITHER The model predicts every high tide 8.55 m. The next high tide 8.91 is higher than that so not perfect model OR Time difference between high tide and low tide is 6 hr 20 minutes, and between low tide and the next high tide is 5 hours and 40 minutes. The model gives these times as equal, so not perfect model OR tide reaches 8.91 m when $\cos(at + b) = 1.105$ which is impossible OR When $t = 12.983$ $h = 8.35$ which is less than the given value of 8.91 m so the model is not suitable	B1(AO3.4) E1(AO3.5b) [2] B1 E1 [2] B1 E1 [2] B1 E1 [2]	Allow for a comment about the maximum height being wrong. FT their values Allow for a comment that the time of the next high tide is wrong. FT their values Allow for a comment that the height predicted cannot reach 8.91 m. FT their values Allow for a comment that the height predicted is not 8.91 m. FT their values	Allow for $t = 13$ but not $t = 12.59$
		Total	10		

Question		Answer/Indicative content	Marks	Guidance
11	a	$y = \frac{1}{7 \cos x - 2 \sin x} = \frac{1}{\sqrt{53}} \sec(x + \alpha)$ In either order Stretch scale factor $\frac{1}{\sqrt{53}}$ in the y-direction Translation $\begin{pmatrix} -0.278 \\ 0 \end{pmatrix}$	B1 (AO 1.1a) B1 (AO 1.1b) B1 (AO 1.1b) [3]	Stretch in the y -direction for correct scale factor FT their value for R FT their value for α. Allow for translation to the left by 0.278 Do not allow enlargement instead of stretch Do not allow for “shift” or “slide” used instead. Examiner's Comments Candidates who wrote out the equation of the curve in R, α in form were generally the most successful here. The descriptions of the transformations were often attempted without technical language or vector notation. It was a predictable error to give the scale factor as $\sqrt{53}$. Follow-through was given for incorrect values of R and α.  Afl Candidates should expect that the two parts of a question like this are likely to be linked and that they should use their answer from part (a) here. They should also be prepared to use the technical language here, i.e. stretch and scale factor rather than enlargement as well as translation instead of shift / move along with proper vector notation used.

Question		Answer/Indicative content	Marks	Guidance
	b	$7 \cos x - 2 \sin x = R \cos(x + \alpha)$ $\Rightarrow 7 = R \cos \alpha, 2 = R \sin \alpha$ $R = \sqrt{7^2 + 2^2} = \sqrt{53}$ $\alpha = \arctan \frac{2}{7} = 0.278\dots$	M1 (AO 1.1a) B1 (AO 1.1b) M1 (AO 1.1a) A1 (AO 1.1b) [4]	Forming two equations soi. Allow sign errors. Allow even if from equations with sin/cos interchange; must be exact FT their equations cao (3sf) Must be in radians for the A mark. <u>Examiner's Comments</u> This is a very standard question that most candidates were at least able to begin. Some solutions were chaotic in how they were set out, however. There was some confusion with signs and which way up the equation for $\tan \alpha$ should be. Marks were lost for decimal values for R as well as values for α incorrectly rounded or given in degrees.
		Total	7	

Question			Answer/Indicative content	Marks	Guidance
12	a		$\theta = \frac{72.576}{r^2}$ or $72.576 r^{-2}$ isw	B1 (AO1.1) [1]	eg $\frac{9072}{125r^2}$ or $\frac{9072r^{-2}}{125}$ <u>Examiner's Comments</u> Candidates who did well in this question worked with the correct formula and simplified their answer. Candidates who did not do well either worked with the wrong formula, usually $\text{area} = r^2\theta$ or $\text{area} = \frac{1}{2}r\theta^2$.
	b		$r\theta + 2r$ or $r(\theta + 2) = 24.48$ seen and $\theta = \frac{24.48 - 2r}{r}$ or equivalent constructive step to give $\frac{24.48}{r} - 2$ AG	B1 (AO1.1) [1]	or $2\pi r \frac{\theta}{2\pi} + 2r = 24.48$ <u>Examiner's Comments</u> Candidates who did well in this question started with a correct formula and supplied a correct intermediate step to derive the given formula. Some candidates attempted to work backwards from the given answer; this should be discouraged since it rarely provides a clear mathematical argument for full credit. Some candidates converted to degrees; whilst in theory this could provide a valid mathematical argument, the majority using this method did not complete the 'show that' request by converting back into radians to confirm the given answer.

Question			Answer/Indicative content	Marks	Guidance	
	c		$their \frac{2 \times 36.288}{r^2} = \frac{24.48}{r} - 2 \text{ oe}$ $r^2 - 12.24r + 36.288 [= 0]$ $[r =] 5.04 \text{ or } 7.2 \text{ oe}$	M1 (AO3.1a) M1 (AO2.1) A1 (AO1.1) [3]	NB 72.576 quadratic obtained in form $f(r) [= 0]$	allow B3 for 5.04 and 7.2 unsupported or allow SC3 for obtaining and solving an equation for θ and then finding both values of r
			Total	5	Examiner's Comments Candidates who did well in this question equated the answers to these parts and manipulated their equation successfully to produce a quadratic, which was then usually solved correctly. Candidates who did less well often knew what approach to adopt but were unable to deal with the algebra successfully.	

Question		Answer/Indicative content	Marks	Guidance	
13	a	$\text{LHS} = \frac{\sin^2 \theta - (1 - \cos \theta)}{(1 - \cos \theta) \sin \theta}$ $= \frac{(1 - \cos^2 \theta) + \cos \theta - 1}{(1 - \cos \theta) \sin \theta}$ $= \frac{\cos \theta (1 - \cos \theta)}{\sin \theta (1 - \cos \theta)}$ $= \cos \theta$ Alternative solution $\frac{\sin \theta}{(1 - \cos \theta)} \cdot \frac{(1 + \cos \theta)}{(1 + \cos \theta)} = \frac{\sin \theta (1 + \cos \theta)}{\sin^2 \theta}$ So LHS becomes $\frac{(1 + \cos \theta)}{\sin \theta} - \frac{1}{\sin \theta} = \frac{\cos \theta}{\sin \theta} = \cot \theta$	M1 (AO 2.1) B1 (AO 2.1) M1 (AO 2.1) A1 (AO 2.1) [4] M1 B1 M1 A1	Attempt to write LHS as a single fraction Use of identity $\sin^2 \theta = 1 - \cos^2 \theta$ Algebraic manipulation eg factorising the numerator AG Complete proof Attempt to change the denominator of the fraction Use of trig identity Combining the fractions AG Complete proof	Where candidates manipulate the entire statement, allow M1 for eliminating or combining fractions eg multiplying through by $(1 - \cos \theta) \sin \theta$ M1 for algebraic manipulation leading to a known identity B1 identity obtained. A1 Complete proof

Examiner's Comments

The majority of good answers were well structured, working from the left-hand side to the right. Many candidates began well, attempting to use a common denominator and replacing $\sin^2 \theta$ with $1 - \cos^2 \theta$, but did not factorise to complete the proof. A significant number of candidates had fragments of working, presumably with restarts, but no wording or structure to help the examiner follow their argument. Candidates who began with the complete statement and rearranged to reach a known identity rarely argued it sufficiently well to get full marks.

Question			Answer/Indicative content	Marks	Guidance
	b		Uses $\frac{1}{\tan \theta} = 3 \tan \theta$ $\tan \theta = \pm \frac{1}{\sqrt{3}}$ $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$	M1 (AO 1.1a) M1 (AO 1.1a) A1 (AO 1.1b) [3]	soi. Also allow for equivalent equation in $\cos \theta$ Method must be clearly using the given answer in (a) Allow positive root only for second M mark Do not allow if additional answers in the interval; ignore additional values outside the interval. Examiner's Comments Most candidates used the given result to obtain a quadratic in $\tan \theta$, a few candidates finding quadratics in $\sin \theta$ or $\cos \theta$. Many only used the positive square root and so only obtained one root. Others lost marks for giving their answer as decimals or in degrees.  AFL Check whether the question expects more than one root and if your working only gives one root, check for an error or omission.
			Total	7	

Question		Answer/Indicative content	Marks	Guidance
14	a	Sector area $= \frac{1}{2}r^2x$ and Triangle $= \frac{1}{2}r^2 \sin x$ Area segment $= \frac{1}{2}r^2 (x - \sin x)$ $\frac{1}{2}r^2 (x - \sin x) = 0.05\pi r^2$ $x - \sin x = 2 \times 0.05 \times \pi \Rightarrow x - \sin x = \frac{1}{10} \pi$	M1 (AO 2.1) A1 (AO 2.1) B1 (AO 2.1) A1 (AO 2.1) [4]	Both areas seen segment area found 0.05πr^2 oe seen AG Must be fully shown and correct rearrangement Examiner's Comments For candidates who clearly labelled the areas of the regions, this was quite a straightforward proof. Errors usually came from omitting the square for the radius or using incorrect formulae. Some candidates looked to the given answer and tried to get the 5% to give the fraction 1/10.
	b	$x_{n+1} = x_n - \frac{x_n - \sin x_n - \frac{1}{10} \pi}{1 - \cos x_n}$	B1 (AO 1.1a) [1]	$x_{n+1} =$ must be seen Algebraic form must be seen Derivative must be worked out Examiner's Comments The generic formula for Newton-Raphson iteration is given in the formula pages, so candidates should not have expected that all that was required was copying this and doing nothing more. Some lost the mark for the formula by giving only the expression from the right-hand side, missing the $x_{n+1} =$ to complete the formula. Where the formula only appeared in part (c), the mark for part (b) was not given retrospectively unless the candidate indicated where their answer was to be found. Condone x used instead of x_n in the fraction part.

Question		Answer/Indicative content	Marks	Guidance	
	c	$x_0 = 1.2$ $x = 1.27245\dots$ $x = 1.26895\dots$ $x = 1.26894\dots$ So root is 1.269 to 3 dp	M1 (AO 1.1a) A1 (AO1.1b) A1 (AO 2.2b) [3]	3 iterations recorded The first 3 iterations correct Allow 1.27, 1.269 or more decimal places if correct.	Root is 1.268947865 to 9 dp Examiner's Comments A calculator set in degrees gave a sequence that did not converge, but many candidates did not realise how to fix the issue. Some candidates who had a converging sequence did not use their answers to give a value for the root. Exemplar 4 $x_0 = 1.2$ $x_{n+1} = \sin x_n + \frac{1}{16} \pi$ $x_1 = 1.24642$ $x_2 = 1.2462$ $x_3 = 1.2668$ $x_4 = 1.2683$ $x_5 = 1.2688$ $x_6 = 1.2689$ $x_7 = 1.2689$ $x_8 = 1.2689$ $x = 1.269$ The question asks for the Newton-

Question			Answer/Indicative content	Marks	Guidance
					Raphson method to be used, yet in this example the $x = g(x)$ method is used. The method mark in part (c) was awarded for three terms of whichever sequence the candidate used but the first accuracy mark was not awarded since these are not the correct terms from the Newton-Raphson method. The second accuracy mark was given here as the sequence is seen converging to the correct root.
			Total	8	
15			$\frac{1}{2} \times 14 \times 8 \sin 117^\circ$ 49.9 [cm ²]	M1 (AO1.2) A1 [2]	Condone omission of $\frac{1}{2}$ Accept 50 50.0 is wrong <u>Examiner's Comments</u> This question was generally done correctly using the area formula, though a small minority of candidates found a value for the height of the triangle first. A small number of candidates forgot the 'half' in the area formula, while just a few candidates used cosine rather than sine. Some candidates did not give the answer correct to 3 significant figures as requested in the question.
			Total	2	

Question		Answer/Indicative content	Marks	Guidance
16	a	$\sin BAC = \sqrt{1 - \cos^2 BAC} \quad \left[= \frac{4\sqrt{14}}{15} \right]$ $\text{Area} = \frac{1}{2} \times 5 \times 9 \times \sin BAC$ $= 6\sqrt{14} \text{ cm}^2$	B1 (AO3.1a) M1 (AO1.1a) A1 (AO1.1) [3]	FT their (a) Use of $\frac{1}{2} \times 5 \times 9 \times \sin 86^\circ$ or similar, using their value for another angle found B0 M1 A0 Allow if value used for their angle Cao. Must be from exact working Condone missing units Examiner's Comments The formula for area was well understood, although a small number of candidates felt the need to evaluate the angle C rather than use $A = \frac{1}{2}bc \sin A$. Most gave a decimal answer using a value for A rather than use the exact link between sin A and cos A.  AFL When the exam question uses the word exact, candidates should be expecting to use a fraction, a surd, a multiple of π or a term like $\ln 2$ etc. Exact answers are seldom found by using the decimal value from the calculator no matter how many decimal places are used.

Question			Answer/Indicative content	Marks	Guidance
	b		$\cos BAC = \frac{5^2 + 9^2 - 10^2}{2 \times 5 \times 9}$ $= \frac{1}{15}$	M1 (AO1.1a) A1 (AO1.1) [2]	Oe Do not allow for a different angle found Fraction must be seen in lowest terms isw 86.2° found <u>Examiner's Comments</u> Most candidates knew the cosine formula but some struggled to rearrange it accurately. Many also went on to find the value of the angle BAC, the fact that they were not asked for it might have led them to realise what was required in part (b).
			Total	5	

Question	Answer/Indicative content	Marks	Guidance
17	DR $4 \sin x^\circ = 3 \cos x^\circ$ $\tan x^\circ = \frac{3}{4}$ $x^\circ = 36.9^\circ, 216.9^\circ$ Alternative method $4 \sin x^\circ = 3 \cos x^\circ$ $\Rightarrow 16 \sin^2 x^\circ = 9 \cos^2 x^\circ$ $\sin^2 x^\circ = \frac{9}{25} \text{ or } \cos^2 x^\circ = \frac{16}{25}$ $x^\circ = 36.9^\circ, 143.1^\circ, 216.9^\circ, 323.9^\circ$ Check solutions satisfy the original equation to give $x^\circ = 36.9^\circ, 216.9^\circ$	M1 (AO1.1a) M1 (AO1.1a) A1 (AO1.1) [3] M1 (AO1.1a) M1 (AO1.1a) A1 (AO1.1) [3]	Attempt to solve simultaneously Using trig identity to form equation in $\tan x$ Both answers required with no extras in the interval. ISW answers outside $0^\circ \leq x^\circ \leq 360^\circ$ Attempt to solve simultaneously Using trig identity $\sin^2 x^\circ + \cos^2 x^\circ = 1$ to form equation in $\sin^2 x^\circ$ or $\cos^2 x^\circ$ Both answers required with no extras in the interval. ISW answers outside $0^\circ \leq x^\circ \leq 360^\circ$ Examiner's Comments Most candidates realised that this requires the two equations to be solved simultaneously and went on to achieve full marks. It was common to see mistakes in the rearranging leading to $\tan x = \frac{4}{3}$ which loses the accuracy mark. Some candidates lost the final mark as they did not check the rounding necessary in this question.  AfL Correct answer can be obtained using a calculator with no working and were awarded no marks. Because this question is labelled as a detailed reasoning

Question			Answer/Indicative content	Marks	Guidance
					question, candidates need to be aware that all the steps of working should be shown to achieve full marks.
			Total	3	
18		a	$-\frac{1}{x^2} \cos\left(\frac{1}{x}\right)$	B2 (AO 1.1a, 1.1) [1]	B1 for $-\frac{1}{x^2}$ or $\cos\left(\frac{1}{x}\right)$
		b	$\frac{1}{x} > 0$ $\frac{1}{x} < \frac{\pi}{2}$	B1 (AO 2.2a) B1 (AO 2.2a) [2]	Correct lower limit 0 stated Correct upper limit $\frac{\pi}{2}$ stated
		c	$0 < \frac{1}{x} < \frac{\pi}{2} \Rightarrow \cos\left(\frac{1}{x}\right) > 0$ $-\frac{1}{x^2} < 0$ Hence $h'(x) < 0$ so $h(x)$ is a decreasing function	M1 (AO 2.2a) M1 (AO 1.1) A1 (AO 2.4) [3]	AG Convincing completion needed
			Total	7	

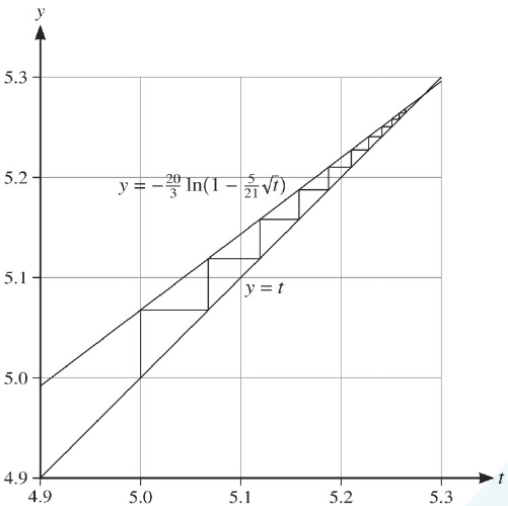
Question			Answer/Indicative content	Marks	Guidance
19		a	$0, 2\pi$ π	B1(AO 1.2) B1(AO 1.2) [2]	
		b	$\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$, so $\frac{2}{y} = 1 + \left(\frac{x}{2}\right)^2$ $\frac{2}{y} = \frac{4 + x^2}{4}$ $y = \frac{8}{x^2 + 4}$ Alternative solution 1 $x^2 = 4 \cot^2 \theta = \frac{4 \cos^2 \theta}{\sin^2 \theta}$ $x^2 = \frac{2(2 - y)}{\frac{1}{2}y}$ $y = \frac{8}{x^2 + 4}$ Alternative solution 2 $\frac{A}{x^2 + B} = \frac{A}{4 \cot^2 \theta + B} = 2 \sin^2 \theta$ $8 \cos^2 \theta + 2B \sin^2 \theta = A$ so $B = 4$ and $A = 8$, giving $y = \frac{8}{x^2 + 4}$	M1(AO 3.1a) M1(AO 1.1) A1(AO 2.2a) M1 M1 A1 M1 M1 A1 [3]	

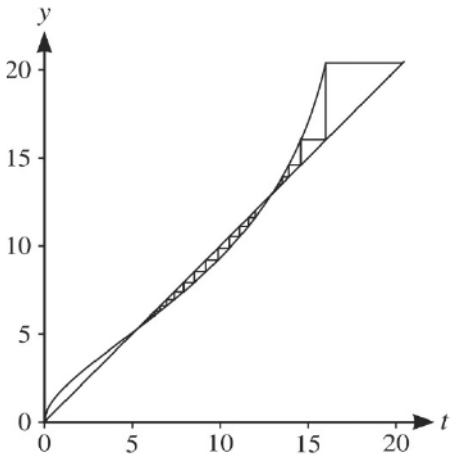
Question			Answer/Indicative content	Marks	Guidance	
		c	$\frac{8}{x^2 + 4} > 0$ for all x So Ali is correct [because $y > 0$ for all x]	M1(AO 2.1) A1(AO 2.3) [2]	or $2\sin^2\theta \geq 0$ and when $2\sin^2\theta = 0$, x is undefined If zero scored, SC1 for $x^2 + 4$ is never negative oe	
			Total	7		
20		a	$a = \frac{1}{2}(17.03 - 7.47) = 4.78$ $c + 4.78 = 17.03$ so $c = 12.25$	B1 (AO 3.1b) B1(AO 3.3) [2]	$\sin\theta = 1$ for max $\sin\theta = -1$ for min B1 $17.03 = a + c$, $7.47 = -a + c$ B1	BC Sufficient reasoning needed to justify given answers
		b	$\frac{2\pi}{365} \times 172 + b = \frac{\pi}{2}$ or $\frac{2\pi}{365} \times 355 + b = \frac{3\pi}{2}$ $b = -1.39$	M1 (AO 3.3) A1(AO 1.1) [2]		
		c	$t = 244$ used in their formula $Y = 13.81$ which is fairly close to 13.75 (out by 3.6 minutes)	M1 (AO 3.4) A1(AO 3.5a) [2]	BC	
		d	$a = 8.51$ and $c = 12.63$	B1 (AO 3.5b) [1]		
		e	New model gives 15.40 hrs, which is not a good fit	B1 (AO 3.5a) [1]	NB 15.39828...	
			Total	8		

Question			Answer/Indicative content	Marks	Guidance
21		a	$[\cos^2 x =] \frac{1}{2}(1 + \cos 2x)$	B1 (AO 1.1a) [1]	
		b	$R = 10$ $\theta = \arctan(0.75)$ isw or 0.643501... to 3 or more sf	B1 (AO 1.1) B1 (AO 1.1) [2]	
		c	DR substitution of results from parts (a) and (b) in the equation $6\sin 2x + 8\cos 2x = 5$ $\arccos\left(\frac{5}{R}\right)$ found FT their R 0.845, 3.99, 2.94, 6.08 cao	M1 (AO 2.1) A1 (AO 1.1) M1 (AO 3.1a) A1 (AO 1.1) A1 (AO 1.1) [5]	if A0A0 allow A1 for all four values correct to a different precision
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
22		a	$12000 = 22000 - a\sqrt{4}$ $a = 5000$	B1 (AO 3.3) [1]	cao
		b	$\frac{dV}{dt} = -5000 \times \frac{1}{2} \times t^{-\frac{1}{2}}$ $t = 4 \Rightarrow \frac{dV}{dt} = -1250$, so the caravan is losing value at a rate of £1250 per year when $t = 4$	M1 (AO 3.4) A1 (AO 1.1b) A1 (AO 3.4) [3]	Attempt to differentiate Correct derivative, in any form Must interpret the negative value of the derivative and must state units
		c	For large values of t , [more than 19.36 years] the value of V is negative which is not possible	E1 (AO 3.5b) [1]	Any argument based on negative values for [sufficiently] large t
		d	$c = 1000$ $22\,000 = be^0 + c$ so $b = 21000$	B1 (AO 3.3) B1 (AO 3.3) [2]	cao cao

Question			Answer/Indicative content	Marks	Guidance	
		e	Agree when $22\,000 - 5\,000\sqrt{t} = 21\,000e^{-0.15t} + 1000$ i.e. when $e^{-0.15t} = \frac{21\,000 - 5000\sqrt{t}}{21\,000} = 1 - \frac{5}{21}\sqrt{t}$ $-0.15t = \ln\left(1 - \frac{5}{21}\sqrt{t}\right)$ $t = -\frac{1}{0.15} \ln\left(1 - \frac{5}{21}\sqrt{t}\right)$ $= -\frac{20}{3} \ln\left(1 - \frac{5}{21}\sqrt{t}\right)$	M1 (AO 2.1) M1 (AO 2.1) A1 (AO 2.1) [3]	Equating their models and attempting to isolate the exponential term Taking logs of both sides AG; complete argument needed	$e^{-0.15t} = K$ need not be simplified at this stage
		f	$t_1 = 5.067572851$ $t_2 = 5.118888519$	B1 (AO 1.1b) B1 (AO 1.1b) [2]	BC Allow answers which agree when rounded to 3sf	

Question			Answer/Indicative content	Marks	Guidance
		g		B1 (AO 1.1b) [1]	Must show at least two steps beginning at $t = 5$
		h	The argument is false – the sequence is increasing and could [and in this case does] increase so that the root rounds to 5.3 or higher	B1 (AO 2.3) [1]	Accept an argument based on calculation of additional terms Do not accept “false” without a reason

Question			Answer/Indicative content	Marks	Guidance
		i	 So the iteration will not find the root near 12 Alternative solution 1 Sequence starting above the root, e.g. 13, 13.035, 13.089, 13.174, 13.310, 13.532, 13.909 and sequence starting below the root, e.g. 12, 11.611, 11.118, 10.530, 9.874, 9.193, 8.532, K So the iteration will not find the root near 12 Alternative solution 2 Near the root, the gradient of $y = -\frac{20}{3} \ln\left(1 - \frac{5}{21}\sqrt{t}\right)$ is greater than 1 So the iteration will not find the root near 12	B1 (AO 2.4) B1 (AO 2.2a) B1 B1 B1 B1 [2]	Sketch showing two staircases, one starting above the root and diverging and one starting below the root and converging on the root near 5 Correct conclusion for correct reason Increasing sequence starting above and decreasing sequence starting below Correct conclusion for correct reason For correct relevant statement about the gradient Correct conclusion for correct reason
			Total	16	

Question			Answer/Indicative content	Marks	Guidance
24			$\cos C = \frac{5^2 + 8^2 - 7^2}{2 \times 5 \times 8} = 0.5$ $C = \frac{1}{3}\pi$ $\text{Area triangle} = \frac{1}{2} \times 5 \times 8 \times \sin \frac{1}{3}\pi$ $\text{Area of sector} = \frac{1}{2} \times r^2 \times \frac{1}{3}\pi = \frac{1}{2} \times \left(\frac{1}{2} \times 5 \times 8 \sin \frac{1}{3}\pi \right)$ $r^2 = \frac{6}{\pi} \times 10 \sin \frac{1}{3}\pi \Rightarrow r = 4.07 \text{ to 3sf}$	M1 (AO 3.1a) A1 (AO 1.1b) B1 (AO 1.1a) M1 (AO 3.1a) A1 (AO 1.1b) [5]	Use of the cosine rule to find angle C Allow 60° May be implied Forming an equation for r^2 Allow 4.1 or better
			Total	5	

Question			Answer/Indicative content	Marks	Guidance	
25		a	Using $\sin \theta \approx \theta$ with $\theta = 2x^2$ and $\cos \theta \approx 1 - \frac{1}{2}\theta^2$ with $\theta = 5x$ gives $2x^2 - \left(1 - \frac{1}{2}(5x)^2\right) = 0$ $\frac{29}{2}x^2 = 1 \Rightarrow x = \sqrt{\frac{2}{29}} = 0.2626\text{K}$ So estimate is 0.26 to 2 decimal places	M1 (AO 2.1) M1 (AO 2.1) A1 (AO 2.1) [3]	Allow slip in $(5x)^2$ Attempt to solve for x AG Must be rounded to 2 dp following either exact answer or answer to more than 2 dp seen	
		b	Denoting $\sin 2x^2 - \cos 5x$ by $f(x)$: $f(0.255) = -0.1618\dots$ and $f(0.265) = -0.1033\dots$ These are both negative so the root does not lie between 0.255 and 0.265, so the estimate is not correct to 2 decimal places	M1 (AO 2.1) A1 (AO 2.2a) [2]	Search for sign change using their 2dp value ± 0.005 Complete argument from correct figs	oe, e.g. comparing values of $\sin 2x^2$ and $\cos 5x$ at end-points
		c	$f'(x) = 4x \cos 2x^2 + 5 \sin 5x$ $x_1 = 0.2 - \frac{\sin(2 \times 0.2^2) - \cos(5 \times 0.2)}{4 \times 0.2 \cos(2 \times 0.2^2) + 5 \sin(5 \times 0.2)}$ $= 0.2 - \frac{-0.4603876119}{5.004796289} = 0.29198928\text{K}$ so $x_1 = 0.2919893$ correct to 7 dp	M1 (AO 1.1a) M1 (AO 1.1a) A1 (AO 1.1) M1 (AO 2.1) A1 (AO 2.1) [5]	Attempt to find $f'(x)$ Use of chain rule for $\sin 2x^2$ Derivative fully correct Use of iterative formula to find x_1 AG Answer must follow from clear working	

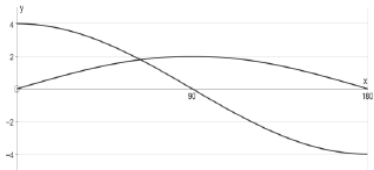
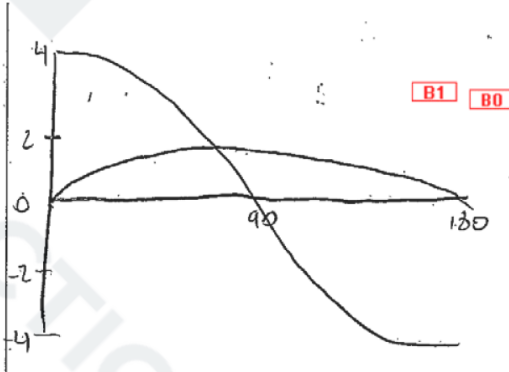
Question			Answer/Indicative content	Marks	Guidance
		d	$x_2 = 0.282\ 338\ 3$ $x_3 = 0.282\ 285\ 3$ Root is 0.282 correct to 3sf	M1 (AO 1.1a) A1 (AO 1.1) B1 (AO 2.2b) [3]	Finds at least one more iteration For correct x_3
			Total	13	
26		a	$(\operatorname{cosec} \theta - \cot \theta)^2 = \left(\frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta} \right)^2$ $= \frac{(1 - \cos \theta)^2}{\sin^2 \theta} = \frac{(1 - \cos \theta)^2}{1 - \cos^2 \theta}$ $= \frac{(1 - \cos \theta)^2}{(1 - \cos \theta)(1 + \cos \theta)} = \frac{1 - \cos \theta}{1 + \cos \theta}$	M1 (AO 2.1) M1 (AO 2.1) A1 (AO 2.1) [3]	Using $\operatorname{cosec} \theta = \frac{1}{\sin \theta}$, $\cot \theta = \frac{\cos \theta}{\sin \theta}$ Using $\sin^2 \theta = 1 - \cos^2 \theta$ AG Factorising must be shown
		b	$\frac{1 - \cos \theta}{1 + \cos \theta} = \frac{1}{3} \Rightarrow 3 - 3 \cos \theta = 1 + \cos \theta \Rightarrow \cos \theta = \frac{1}{2}$ $\theta = 60^\circ, 300^\circ$	M1 (AO 1.1a) A1 (AO 1.1) A1 (AO 1.1) [3]	Attempt to rearrange and find $\cos \theta$ For one correct value for θ For second correct value; do not allow if additional values in range given, but ignore values outside range
			Total	6	

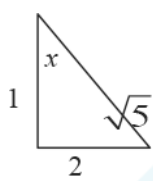
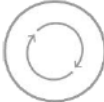
Question			Answer/Indicative content	Marks	Guidance
27		a	$R = \sqrt{3}$ $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ $\alpha = 0.615$	B1 (AO1.1) M1 (AO1.1) A1 (AO1.1) [3]	0.61547970... rounded to 2 or more significant figures
		b	$f(x) = \frac{5}{2 + \sqrt{3} \cos(x + 0.62)}$ At min value, $\cos(x + 0.62) = 1$ soi $10 - 5\sqrt{3}$	M1 (AO3.1a) M1 (AO2.1) A1 (AO1.1) [3]	FT their R BC rationalising
			Total	6	
28			$\cos \theta = \frac{9^2 + 5^2 - 12^2}{2 \times 5 \times 9}$ $-\frac{38}{90}$ oe soi 115°	M1 (AO3.1) A1 (AO1.1) A1 (AO1.1) [3]	May be implied by 114.97... degrees or 2.0066917... radians
			Total	3	

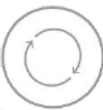
Question			Answer/Indicative content	Marks	Guidance	
29		a	x is not defined for this value	E1 (AO 2.4) [1]		
		b	$\cos 2\theta$ has max value 1 when $\theta = 0$ so max y is 2	E1 (AO 2.4) [1]	Or using $\frac{dy}{d\theta} = 0$ correctly	
		c	$y = 2 \cos^2 \theta$ $1 + x^2 = \sec^2 \theta$ So $1 + x^2 = \frac{2}{y} \Rightarrow y = \frac{2}{1 + x^2}$	M1 (AO 2.2a) M1 (AO 1.1a) E1 (AO 1.1) [3]	Relevant use of $\cos 2\theta = 2 \cos^2 \theta - 1$ Relevant use of $\sec 2\theta = 1 + \tan^2 \theta$ AG Successful completion	At any stage At any stage With use of $\sec \theta = \frac{1}{\cos \theta}$

Question			Answer/Indicative content	Marks	Guidance	
		d	DR $\frac{dy}{dx} = \frac{-4x}{(1+x^2)^2}$ Gradient OP = $\frac{y}{x}$ $\frac{-4x}{(1+x^2)^2} \times \frac{y}{x} = -1$ $\frac{4x}{(1+x^2)^2} \times \frac{2}{x(1+x^2)} = 1 \Rightarrow (1+x^2)^3 = 8 \Rightarrow 1+x^2 = 2$ $x = 1$ Point P is (1, 1)	M1 (AO 1.1) A1 (AO 1.1) B1 (AO 3.1a) M1 (AO 2.2a) M1 (AO 2.1) A1 (AO 2.2a) A1 (AO 3.2a) [7]	Use of chain rule or quotient rule Correct derivative May be implied Form and solve equation for x Allow ± 1 at this stage	Or, using parameters: $\frac{dy}{dx} = \frac{dy}{d\theta} \bigg/ \frac{dx}{d\theta} = \frac{-2 \sin 2\theta}{\sec^2 \theta}$ $\frac{-2 \sin 2\theta}{\sec^2 \theta} \frac{(1 + \cos 2\theta)}{\tan \theta} = -1$ $\frac{4 \sin \theta \cos^3 \theta}{1} \times \frac{2 \cos^3 \theta}{\sin \theta} = 1$ $\cos \theta = (\pm) \frac{1}{\sqrt{2}}$
			Total	12		

Question			Answer/Indicative content	Marks	Guidance	
30		a	$\frac{1}{2} \times 2^2 \times \theta = \frac{2}{3} \pi$ $\theta = \frac{1}{3} \pi$ $x = 2 \cos \frac{1}{3} \pi, \quad y = 2 \sin \frac{1}{3} \pi$ $(1, \sqrt{3})$	M1 (AO 3.1a) A1 (AO 1.1) M1 (AO 1.1) A1 (AO 2.2a) [4]	Use of $r = 2$ in sector formula Use of trig for either coordinate Coordinates must be exact	Allow separate $x =$, $y =$
		b	Midpoint of AB is $(\frac{3}{2}, \frac{1}{2}\sqrt{3})$ Centre of new circle is $(3, \sqrt{3})$ Equation is $(x-3)^2 + (y-\sqrt{3})^2 = 4$	M1 (AO 1.1) M1 (AO 3.1a) A1 (AO 1.1) [3]	FT <i>their</i> B throughout part (b) oe	Or use of $\begin{pmatrix} 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix}$ $\sqrt{\quad}$
			Total	7		

Question	Answer/Indicative content	Marks	Guidance
31	a $y = 4\cos x$ $y = 2\sin x$ 	B1 (AO1.2) B1 (AO1.2) [2]	Correct shapes and relative sizes Graphs labelled with correct amplitudes clear Ignore any graph beyond the given interval Examiner's Comments The graphs were generally well done although many candidates lost a mark as they did not indicate which graph was which. Some just drew the basic graphs of $y = \sin x$ and $y = \cos x$ with an amplitude of 1. Only rarely did candidates draw the graph of $y = \cos 4x$. Some candidates who used their calculators to produce points to plot $y = 4\cos x$ gave too few points and drew a line from (0, 4) to (180, -4)  Make sure all graphs are fully labelled. Exemplar 1  B1 B0

Question	Answer/Indicative content	Marks	Guidance
	b $4\cos x = 2\sin x$ $\tan x = 2$ $x = \arctan 2$ $4\cos x = 4 \times \frac{1}{\sqrt{5}} = \frac{4}{\sqrt{5}}$  Point of intersection is $\left(\arctan 2, \frac{4}{\sqrt{5}}\right)$	M1 (AO3.1a) A1 (AO1.1b) M1 (AO1.1a) A1 (AO1.1b) [4]	Forming equation leading to $\tan x = k$ oe Award for 63.4° for this first A mark Attempting to find exact value for $\cos x$ or $\sin x$ Must be a coordinate in this form <u>Examiner's Comments</u> Many candidates divided by $\cos x$ to obtain an equation for $\tan x$, but there were many who got mixed up and obtained $\tan x = \frac{1}{2}$ or even $\tan x = 0$. Most did not think to use Pythagoras' theorem to find the y coordinate of the point of intersection.  Where a value for $\tan x$ is known, draw a simple right angled triangle and using Pythagoras' theorem to give the values for $\sin x$ or $\cos x$. Do not allow for using 63.4°

Question			Answer/Indicative content	Marks	Guidance
		c	The student's argument is invalid as the period of $\tan x$ is 180° .	E1 (AO2.3) [1]	Must state or imply that the argument is invalid and give a correct reason Allow for "roots occur every 180°" oe Examiner's Comments Few candidates used the periodicity of $\tan x$ to answer this question but many correct answers explained that there would be another point of intersection for $180^\circ \leq x \leq 360^\circ$.  Make sure that you clearly state whether the argument is valid or not and make sure this does not contradict your evidence.
			Total	7	

Question			Answer/Indicative content	Marks	Guidance	
32		a	Angle BDC = $90 - \theta$ (angles of triangle) Angle CDA = θ (Angle ADB = 90° as it is the angle in a semicircle)	M1 (AO 2.1) E1 (AO 2.2a) [2]	Including reason Answer given so mark is for reason Examiner's Comments This question drew on prior knowledge from GCSE. Only few managed to give both 'angles in a triangle' and 'angle in a semicircle'.	Reasons can be given in either order
		b	Triangle ACD, $\tan \theta = \frac{h}{b}$ Triangle BCD, $\tan \theta = \frac{a}{h}$ $\frac{a}{h} = \frac{h}{b} \Rightarrow h^2 = ab \Rightarrow h = \sqrt{ab}$	M1 (AO 1.1) E1 (AO 2.1) [2]	At least one correct expression for $\tan \theta$ Setting expressions equal and correct completion to given answer AG Examiner's Comments This was generally proved correctly.	Alternative method: triangle ACD is similar to triangle DBC
			Total	4		

Question			Answer/Indicative content	Marks	Guidance
33			$AC = [AO - OC] = 1 - \cos\theta$ or $\cos\theta = 1 - AC$	M1(AO 1.1a)	AG Allow $AC = AO - OC$ with $OC = \cos\theta$ for M1
			θ small so $AC \approx 1 - \left(1 - \frac{\theta^2}{2}\right) = \frac{\theta^2}{2}$	E1(AO 2.1)	Convincing completion
				[2]	<u>Examiner's Comments</u> This is the first example of a 'Show that' question in this paper and candidates could not score unless they explained the logic of their initial expression Exemplar 1 $OA = 1$ $OC = \cos\theta = 1 - \frac{\theta^2}{2}$ $1 - 1 + \frac{\theta^2}{2} = CA$ $\frac{\theta^2}{2} = AC$ The candidate knows the small angle approximation to use but does not explain where their first equation for CA comes from. Therefore no marks can be earned.
Total				2	

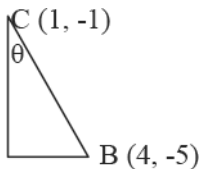
Question			Answer/Indicative content	Marks	Guidance
34			$[BC^2] = 32^2 + 14^2 - 2 \times 32 \times 14 \cos 85^\circ$ $[BC] = 33.8 \text{ cm}$ Perimeter = 79.8cm	M1(AO 3.1a) A1(AO 1.1) A1(AO 1.1) [3]	Use of cosine rule to find BC 1141.9... 33.79... Accept 80 provided 3 or more sf seen as BC Examiner's Comments For the majority of candidates this provided a straightforward start to the paper. The few candidates who did not score full marks either misread the question and only found the length of BC, did not give sufficient accuracy in their answer. A small minority of candidates did not recall accurately the Cosine Rule.  AFL This question is in degrees, but many questions at A Level involve the use of radians. It is important that candidates are confident switching between units on their calculators. The specification advice to explicitly write down any expressions to be evaluated by calculator would ensure partial credit where the incorrect setting on the calculator is used.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance	
35		a	$R = 25$ $\tan^{-1}\left(\frac{24}{7}\right)$ or $\sin^{-1}\left(\frac{24}{25}\right)$ or $\cos^{-1}\left(\frac{7}{25}\right)$ $25\cos(x + 1.29)$	B1 (AO 1.1) M1 (AO 1.1) A1 (AO 1.1) [3]	73.739795° rounded to 2 or more sf may imply M1A0 $\alpha = 1.28700221759$ rounded to 2 or more sf allow A1 for α found to 2 or more sf <u>Examiner's Comments</u> The majority of candidates gained full credit, with careless arithmetic resulting in dropped accuracy marks on this routine item.	
		b	$12 \pm \text{their } 25$ $-13 \leq f(x) \leq 37$	M1 (AO 3.1a) A1 (AO 1.1) [2]	or one of – 13 and 37 identified allow eg from – 13 to 37 inclusive A0 if inequality is strict <u>Examiner's Comments</u> Some candidates answered their own question, taken directly from part (a).	
			Total	5		

Question			Answer/Indicative content	Marks	Guidance	
36		a	C is (1, -1)	B1 (AO 1.1a) [1]	cao	Examiner's Comments Most candidates were correct here.
		b	(i) EITHER Substitute $y = \frac{3}{4}x - 8$ into the equation of the circle $(x-1)^2 + \left(\frac{3}{4}x - 8 + 1\right)^2 = 25$ $x^2 = 8x + 16 = 0$ EITHER $(x-4)^2 = 0$ OR $\text{Discriminant} = (-8)^2 - 4 \times 1 \times 16 = 0$ So the equation has a repeated root so the line is a tangent OR Substitute $x = \frac{4}{3}y - \frac{32}{3}$ into the equation of the circle $\left(\frac{4}{3}y - \frac{32}{3} - 1\right)^2 + (y+1)^2 = 25$ $y^2 + 10y + 25 = 0$	M1 (AO 3.1a) M1 (AO 1.1a) A1 (AO 1.1b) A1 (AO 2.2a) [4] M1 (AO 3.1a) M1 (AO 1.1a) A1 (AO 1.1b)	AG Attempt to eliminate one variable Attempt to expand and collect terms to obtain 3 term quadratic expression A correct 3 term quadratic Clearly argued AG Attempt to eliminate one variable Attempt to expand and collect terms to obtain 3 term quadratic expression A correct 3 term quadratic	

Question			Answer/Indicative content	Marks	Guidance	
			EITHER $(y + 5)^2 = 0$ OR Discriminant = $10^2 - 4 \times 1 \times 25 = 0$ So the equation has a repeated root so the line is a tangent (ii) $x = 4$ and $y = -5$ so B is (4, -5)	A1 (AO 2.2a) [4] B1 (AO 1.1a) [1]	Clearly argued Cao Examiner's Comments There was quite a lot of confusion in this question between the tangent to the circle at A (whose equation is not given nor required) and the line $4y = 3x - 3$ which can be proved to be the tangent to the circle at B. Some candidates set out to find the equation of the tangent at A hoping to obtain the given equation. Successful candidates solved the equation of the line and the circle simultaneously although many struggled with the fractions that resulted. Examiner's Comments There was only one mark here as the x-coordinate had normally been found in part A, so there was only the y-coordinate to find.	

Question			Answer/Indicative content	Marks	Guidance	
		c	$\angle CAD = \angle CBD = 90^\circ$ (radius is perpendicular to the tangent) Gradient of AC $= \frac{2 - (-1)}{5 - 1} = \frac{3}{4}$ Gradient of BC $= \frac{(-1) - (-5)}{1 - 4} = -\frac{4}{3}$ So AC is perpendicular to BC so $\angle ACB = 90^\circ$ So ADBC is a rectangle Either AC = BC radius [=5] Or AD = BD equal tangents so ADBC is a square. $\angle CAD = \angle CBD = 90^\circ$ (radius is perpendicular to the tangent) Gradient of AC $= \frac{2 - (-1)}{5 - 1} = \frac{3}{4}$ Gradient of BD is $\frac{3}{4}$ So AC is perpendicular to BD so ADBE is a rectangle AC = BC radius so ADBC is a square. $\angle CAD = 90^\circ$ (radius is perpendicular to the tangent) AC = BC radius [=5] Gradient of AC $= \frac{2 - (-1)}{5 - 1} = \frac{3}{4}$	B1 (AO 2.1) B1 (AO 3.1a) B1 (AO 2.1) [3] B1 B1 B1 B1	Allow for one or other of these angles Complete proof AG Allow for one or other of these angles Complete proof AG	Allow up to B1, B1 for any two of these three pieces of evidence. Allow the final B1 only when the proof is complete and clearly argued.

Question			Answer/Indicative content	Marks	Guidance	
			Equation of AD is $y - 2 = -\frac{4}{3}(x - 5)$ So coordinates of D are (8, -2) Hence BD = 5 and AD = 5 So ABCD is a rhombus	B1	Gradient of AD must be found from the coordinates of A and C Complete proof Examiner's Comments This proved quite challenging for many candidates. The first 2 marks were given for two facts that form part of a complete proof, and the third only when a complete proof was seen. Some candidates assumed that AD and BD were perpendicular in order to find the equation of AD and subsequently used their equation to complete their answer – but this is not a proper proof. Candidates needed to show they had used the gradient of AC or the derivative of the circle to find the gradient of the tangent at A.	
		d	E is the point (1, -6) EITHER  $\theta = \arctan\left(\frac{3}{4}\right) = 0.6435$ OR $BE = \sqrt{(4-1)^2 + (-5-(-6))^2} = \sqrt{10}$ Cosine rule in triangle BCE $\cos BCE = \frac{5^2 + 5^2 - 10}{2 \times 5 \times 5} \left[= \frac{40}{50} \right]$	B1 (AO2.1) M1 (AO 3.1a) A1 (AO3.1a) (M1)	May be implied Right-angled triangle formed and use of arctan oe Using distance BC and the cosine rule	

Question			Answer/Indicative content	Marks	Guidance
			$\angle BCE = 0.6435\dots$ OR M is the midpoint of BE M is (2.5, -5.5) $BM = \sqrt{(4-2.5)^2 + (-5-(-5.5))^2} = \frac{1}{2}\sqrt{10}$ $\angle BCM = \arcsin\left(\frac{\frac{1}{2}\sqrt{10}}{5}\right) = 0.32175\dots$ $\angle BCE = 0.6435\dots$ Area sector = $\frac{1}{2}r^2\theta = \frac{1}{2} \times 25 \times 2 \times 0.32175\dots$ = 8.04376	(A1) (M1) (A1) M1dep (AO1.1a) A1 (AO1.1b) [5]	oe Using trig in triangle BCM or ECM Allow for $\angle BCM$ Oe. Must be $\angle BCE$ Using the sector area formula FT their $\angle BCM$ <u>Examiner's Comments</u> The problem solving aspect of this question involved the need to find the angle ECB before the formula for the area of the sector could be used. Many candidates managed to write down the coordinates of E for the first mark. There were several valid methods of finding the required angle but it was often difficult to follow candidates thinking unless the correct answer was obtained. The second method mark was dependent on the first so that it was not enough just to guess an angle and use that.  AfL Write a few words of explanation to the examiner, or clear headings, indicating your method so that you can be given a method mark even if your answer is not quite right. Two method marks and the follow-through answer mark may depend on it.
			Total	14	

Question			Answer/Indicative content	Marks	Guidance	
37		a	EITHER $8\sin^2 x \cos^2 x = 2(1 - \cos 2x)(1 + \cos 2x)$ $= 2(1 - \cos^2 2x) = 2 - (1 + 2 \cos 4x)$ $= 1 - \cos 4x$ OR $8\sin^2 x \cos^2 x = 2(2\sin x \cos x)^2$ $= 2\sin^2 2x$ $[= 1 - \cos 2(2x)] = 1 - \cos 4x$ OR $1 - \cos 4x = 1 - (1 - 2\sin^2 2x)$ $= 2\sin^2 2x$ $= 2(2\sin x \cos x)^2$ $= 8\sin^2 x \cos^2 x$	M1 (AO 3.1a) M1 (AO 3.1a) E1 (AO 2.1) [3] M1 M1 E1 [3] M1 M1 E1 [3]	AG Using a double angle formula Second use of a double angle formula Clearly shown Using a double angle formula Another use of a double angle formula Clearly shown Using a double angle formula Another use of a double angle formula Clearly shown	Allow any other valid sequence of identities used. Examiner's Comments This question posed considerable difficulty to many candidates. Some had success by writing both $\sin^2 x$ and $\cos^2 x$ in terms of $\cos 2x$ and multiplying out. Many then did not go on to complete the proof. The first 2 marks were credited for any two applications of the double angle formulae and the final mark only when there was a convincing proof so many candidates were credited 2 out of 3 marks.

Question			Answer/Indicative content	Marks	Guidance
		b	$\int \sin^2 x \cos^2 x \, dx = \frac{1}{8} \int 1 - \cos 4x \, dx$ $= \frac{1}{8}x - \frac{1}{32}\sin 4x + c$	M1 (AO 1.1a) A1 (AO 1.1b) A1 (AO 1.1b) [3]	Attempt to integrate both terms $\frac{1}{4}\sin 4x$ seen or implied All correct. Must include +c Examiner's Comments Most candidates realised they needed to use the identity from part (a) although many omitted the factor of $\frac{1}{8}$. Some candidates lost the final mark for omitting the +c.
			Total	6	

Question		Answer/Indicative content	Marks	Guidance	
38		$(1 + \tan^2 \theta) + 2 \tan \theta = 4$ $\tan^2 \theta + 2 \tan \theta - 3 = 0$ $(\tan \theta - 1)(\tan \theta + 3) = 0$ When $\tan \theta - 1$, $\theta = 45^\circ, 225^\circ$ When $\tan \theta = -3$, $\theta = 108.4^\circ, 288.4^\circ$	M1 (AO 3.1a) M1 (AO 1.1a) A1 (AO 1.1b) A1 (AO 1.1b) [4]	DR Using appropriate trig identity Showing algebraic method for solving their quadratic Any two correct values for θ All correct values for θ and no extras in the interval. Ignore values outside the required interval.	Must attempt to reach an equation with only one trig function eg $20\cos^4 \theta - 12\cos^2 \theta + 1 = 0$ Or $\sqrt{5} \sin(2\theta - 63.4^\circ) = 1$
		Total	4	<u>Examiner's Comments</u> Candidates who used the identity $\sec^2 \theta = 1 + \tan^2 \theta$ generally went on to obtain most of the marks. Only a few candidates tried to rewrite the equation in terms of $\cos \theta$ as this is a much more difficult method requiring both sides to be squared and spurious solutions eliminated. Candidates did not get far enough into this method to obtain the method mark.	

Question			Answer/Indicative content	Marks	Guidance
39		a	$A = 500$ $k = 1.044$	B1(AO3.3) B1(AO1.1) [2]	Examiner's Comments While many candidates wrote down the correct answers from the information given, quite a large minority wrongly calculated the value of k that would make the population 650 after 10 days.
		b	500×1.044^{10} $= 769.1$ which is not close to 650 so not consistent	M1(AO3.4) A1(AO1.1) [2]	Ft www May use 650 and show k or A different Examiner's Comments Candidates who had used the value of 650 to find their value of k could gain only partial credit for this part; the common wrong value of k suggests that the model is consistent. Many candidates correctly found that Munirah's first model would give about 769 flies after 10 days, which is not consistent with 650.
		c	exponential growth oe	B1(AO3.5a) [1]	OR increase for ever, oe Examiner's Comments Almost all candidates realised that the number of flies would continue to rise in the long run.

Question			Answer/Indicative content	Marks	Guidance	
		d	$t = 0, N = 500$ $t = 10, N = 650.37 \approx 650$ model predicts number of fruit flies tends to 750 in the long run	B1(AO1.1) B1(AO1.1) B1(AO3.5a) [3]	Oe 'N will not go beyond 750'	Allow shown using large value of t
Examiner's Comments Although there were many completely correct answers, a lot of candidates identified only one or two of the ways. Some of these candidates used the increase of 4.4% per day from Munirah's original model to suggest an answer; this is not valid. Other candidates made far more general comments that did not answer the question.						

Question			Answer/Indicative content	Marks	Guidance
		e	(i) Oscillations	B1(AO3.5b) [1]	oe <u>Examiner's Comments</u> Very few candidates identified the required feature as (decreasing) oscillations. A considerable number of candidates did not attempt this question.
			(ii) cosine or sine oe	B1(AO3.5c) [1]	Eg Introducing a multiplier of $(-1)^t$ if t is a large integer. <u>Examiner's Comments</u> The most common response was to omit this part. The incorrect answer of a logarithmic function was considerably more common than a correct answer; correct answers invariably involved mention of a sine or cosine function.
			Total	10	

Question			Answer/Indicative content	Marks	Guidance
40		a	DR $\frac{dy}{dx} = 0$ when $\frac{dy}{d\theta} = \cos \theta = 0$ $\theta = \frac{1}{2}\pi, \frac{3}{2}\pi$ $x = -2$ $y = 3$ Alternative solution Maximum y occurs when $\sin \theta = 1$ $y = 3$ $\theta = \frac{1}{2}\pi$ $x = -2$	M1(AO3.1a) M1(AO1.1) A1(AO1.1) A1(AO3.2a) M1 A1 M1 A1 [4]	



Question			Answer/Indicative content	Marks	Guidance
		b	DR $7 \cos \theta + 2 \cos 2\theta = 0 \Rightarrow 4 \cos^2 \theta + 7 \cos \theta - 2 = 0$ $(4 \cos \theta - 1)(\cos \theta + 2) = 0$ $\cos \theta = \frac{1}{4} \text{ or } -2$ Reject $\cos \theta = -2$ $\sin^2 \theta = 1 - \cos^2 \theta = 1 - \left(\frac{1}{4}\right)^2 = \frac{15}{16}$ $y = 2 \pm \frac{\sqrt{15}}{4}$	M1(AO1.1a) M1(AO1.1) A1(AO1.1) B1(AO3.2a) M1(AO3.1a) A1(AO2.2a) [6]	Use of double angle formula Method for solving quadratic May be seen later oe exact form
			Total	10	

Question			Answer/Indicative content	Marks	Guidance
41		a	Tangent perpendicular to radius	B1(AO2.4) [1]	
		b	Draw horizontal line from B to AD: $\cos A = \frac{6-2}{6+2}$ $\cos A = \frac{1}{2} \Rightarrow A = \frac{1}{3}\pi$	M1(AO3.1a) A1(AO2.1a) [2]	AG Successful completion
		c	$DC = (6+2) \sin \frac{1}{3}\pi$ $= 4\sqrt{3}$ Trapezium ABCD has area $\frac{1}{2}(6+2) \times 4\sqrt{3}$ $= 16\sqrt{3} \text{ cm}^2$	M1(AO3.1a) A1(AO1.1) M1(AO1.1) A1(AO2.1) [4]	AG
		d	Angle ABC = $\frac{2}{3}\pi$ $16\sqrt{3} - \frac{1}{2} \times 6^2 \times \frac{1}{3}\pi - \frac{1}{2} \times 2^2 \times \frac{2}{3}\pi$ 4.674 cm ²	M1(AO3.1a) M1(AO1.1) A1(AO1.1) [3]	Either sector correct
			Total	10	

Question			Answer/Indicative content	Marks	Guidance
42		a	DR At intersection $\cos x = \sin x$ $\tan x = 1$ $x = \frac{1}{4}\pi, -\frac{3}{4}\pi$ Between intersections $\cos x - \sin x > 0$, so area = $\int_{-\frac{3}{4}\pi}^{\frac{1}{4}\pi} (\cos x - \sin x) dx$	M1(AO3.1a) M1(AO1.1) A1(AO1.1) E1(AO2.4) [4]	Both values Explanation and completion
		b	DR $[\sin x + \cos x]_{-\frac{3}{4}\pi}^{\frac{1}{4}\pi}$ $\sin\left(\frac{1}{4}\pi\right) - \sin\left(-\frac{3}{4}\pi\right) + \cos\left(\frac{1}{4}\pi\right) - \cos\left(-\frac{3}{4}\pi\right) = \frac{4}{\sqrt{2}}$ $2\sqrt{2}$	M1(AO1.1a) M1(AO2.1) A1(AO1.1) [3]	Limits not needed for method mark
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
43		a	DR $8 \cos x + 5 \sin x = R(\cos x \cos \alpha + \sin x \sin \alpha)$, so $8 = R \cos \alpha$ and $5 = R \sin \alpha$ $R = \sqrt{8^2 + 5^2} = \sqrt{89}$ $\alpha = \arctan\left(\frac{5}{8}\right)$ $8 \cos x + 5 \sin x = \sqrt{89} \cos\left(x - \arctan\left(\frac{5}{8}\right)\right)$	M1(AO1.1a) B1(AO1.1b) A1(AO1.1b) [3]	Equating coefficients Accept 9.43 or better Accept 0.559 or better (No penalty for omission of this step)
		b	DR $\cos\left(x - \arctan\left(\frac{5}{8}\right)\right) = \frac{6}{\sqrt{89}}$, so $x - \arctan\left(\frac{5}{8}\right) = 0.88149\dots$ or $2\pi - 0.88149\dots$ $x = 1.4401$ $x = 5.9603$	M1(AO1.1a) A1(AO1.1a) A1(AO1.1a) [3]	Method leading to at least one solution If a rounded value from (a) used max. A1 only
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
44		a	$AB^2 = 2a^2 - 2a^2 \cos \theta$	B1(AO1.1b) [1]	
		b	$AD = a \sin\left(\frac{1}{2}\theta\right)$ $AB = 2AD \Rightarrow AB^2 = 4AD^2$, so $2a^2 - 2a^2 \cos \theta = 4\left(a \sin\left(\frac{1}{2}\theta\right)\right)^2$ $1 - \cos \theta = 2 \sin^2\left(\frac{1}{2}\theta\right) \Rightarrow \cos \theta = 1 - 2 \sin^2\left(\frac{1}{2}\theta\right)$	B1(AO3.1a) M1(AO2.1) E1(AO2.1) [3]	Must be in terms of a and α Must handle squared term correctly AG Result must be clearly shown
		c	Proved for $0 < \theta < 180^\circ$ (as θ is angle in a triangle)	E1(AO2.3) [1]	Allow 'between' in words Or $0 < \theta < \pi$
			Total	5	

Question			Answer/Indicative content	Marks	Guidance
45			$4 \tan \theta \tan 2\theta = 1 \Rightarrow 4 \tan \theta \cdot \frac{2 \tan \theta}{1 - \tan^2 \theta} = 1$ $\Rightarrow 8 \tan^2 \theta = 1 - \tan^2 \theta$ $\Rightarrow \tan^2 \theta = 1/9$ $\tan \theta = 1/3$ or $-1/3$ $\theta = 18.43^\circ$ or 161.57° $\theta = 18.43^\circ$ and 161.57°	M1* M1dep* A1 A1 [4]	Use of double angle formula for tan to get an equation in tan – allow one sign slip only Re-arranges to $\tan^2 \theta = k$ where $k > 0$ or attempt to solve $a \tan^2 \theta - b = 0$ where $b/a > 0$ One correct answer to at least 1dp Both answers correct to at least 1dp SC A1A0 for answers which round to 0.322 and 2.82 (radians) Answers with no working can score B1 B1 (max 2/4) if correct Ignore additional solutions outside the range. If any additional solutions given inside the range of $0 < \theta < 180$ and full marks would have been awarded then remove last mark (so 3/4) Examiner's Comments It was pleasing to note that most candidates used the correct double angle formulae for $\tan 2\theta$ to obtain a correct equation in terms of $\tan \theta$. However, some candidates over complicated the problem by re-writing tan in terms of sin and cos and in these cases it was extremely rare for candidates to make any real significant progress. Of those that correctly re-arranged $4 \tan \theta \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right) = 1 \text{ to } \tan^2 \theta = \frac{1}{9}$ it was disappointing that so many candidates then only considered the solutions of the equation $\tan \theta = \frac{1}{3}$ and ignored any possible solutions that would have come from the equation $\tan \theta = -\frac{1}{3}$.
			Total	4	

Question			Answer/Indicative content	Marks	Guidance	
46		i	$[AE^2 =] 32^2 + 15^2 - 2 \times 32 \times 15 \times \cos 116$ AE = 40.86...to two or more s.f. isw	M1 A1 [2]	NB 1669.836301 implies M1 Examiner's Comments This was very well done. A few candidates worked in radians and lost the accuracy mark. A small minority misquoted the Cosine Rule or mis-used Pythagoras.	NB 2181.72...or 46.709 implies M1 (radians)
		ii	$\frac{\sin A}{15} = \frac{\sin 116}{\text{their } 40.86}$ or $\frac{\sin E}{32} = \frac{\sin 116}{\text{their } 40.86}$ $h = 32 \times \text{their } \sin A \text{ or } 15 \times \text{their } \sin E$ $h = 10.5 \text{ to } 10.6 \text{ isw}$ <i>Alternatively</i> $\frac{1}{2} \times 32 \times 15 \times \sin 116 = \frac{1}{2} \times \text{their } 40.86 \times h$ $h = \frac{32 \times 15 \times \sin 116}{\text{their } 40.86}$ $h = 10.5 \text{ to } 10.6 \text{ isw}$	M1* M1dep* A1 M1 M1 A1 [3]	$\cos A = \frac{32^2 + \text{their } 40.86^2 - 15^2}{2 \times 32 \times \text{their } 40.86}$ or $\cos E = \frac{15^2 + \text{their } 40.86^2 - 32^2}{2 \times 15 \times \text{their } 40.86}$ $\text{or } \sqrt{32^2 - \text{their } AX^2} \text{ or } \sqrt{15^2 - \text{their } EX^2}$	A = 19.3 and E = 44.7 X is the foot of the perpendicular from D to AE NB 30.2 and 10.7 Examiner's Comments Over half of the candidates failed to score on this part. Most worked with a perpendicular from D to AE and presumed that by doing so they were either bisecting angle ADE or the length AE. Those who correctly worked with the Sine Rule to find angle DAE or angle DEA generally went on to score full marks, although a few found the base of their triangle instead of the height.

Question			Answer/Indicative content	Marks	Guidance	
		iii	$\frac{116}{360} \times \pi \times 10^2$ 101 or 101.2 to 101.23 $\frac{1}{2} \times 32 \times 15 \times \sin 116$ soi 114 to 115 [m ²]	M1 A1 M1 A1 [4]	or $\frac{1}{2} \times 10^2 \times \frac{29\pi}{45}$ oe NB $\frac{29\pi}{45} = 2.02458...$ M0 for $\frac{1}{2} \times 10^2 \times 116$ or $\frac{1}{2} \times \text{their AE} \times \text{their } h$; may be implied by 215.7 to 216	Examiner's Comments Most candidates knew what to do here and successfully found the area of the triangle and the area of the sector. A minority left it at that or slipped up with the subtraction and lost an easy mark. A few candidates used $\theta = 116$ radians, thus losing the first two marks, or converted to radians and then worked with their rounded decimal value, thus losing the accuracy mark.

Question			Answer/Indicative content	Marks	Guidance	
		iv	$\tan 26 = \frac{x}{80}$ or $\tan 64 = \frac{80}{x}$ or $\frac{x}{\sin 26} = \frac{80}{\sin 64}$ oe soi (area of field =) $80 \times 32 + \frac{1}{2} \times 80 \times \text{their } 39.0$ or $\frac{80}{2} [32 + (32 + \text{their } 39.0)]$ 4120 to 4121 area of ADE is 5.2 to 5.24% isw of area of ADCB	M1 M1 A1 B1 [4]	(x is length CF where F is foot of perpendicular from D to BC or length DG where G is foot of perpendicular from C to AD produced) NB $x = 39(.0186070853)$ or $BC = 71.(0\dots)$ may imply M1 or $80 \times [32 + \text{their } 39.0] - \frac{1}{2} \times 80 \times \text{their } 39.0$ NB 4120.74428341 or area of AECB is 94.76 to 94.8% isw of area ADCB	<i>alternatively</i> B3 for (area AEH) awrt 260 and (area HECB) 3640 – 3650 where H is the foot of the perpendicular from E to AB, or B2 for one of these <i>Alternatively</i> B3 for (area AEC) awrt 1060 and (area ABC) awrt 2840 or B2 for one of these allow B3 for 4120 to 4121 not from wrong working or $3905 > 3709$ (area of car park > 90% of field) Examiner's Comments A significant number of candidates were unable to marshal the information to form a coherent strategy for solving this problem, and thus failed to score. A wide variety of approaches were seen, with many opting for convoluted methods which were often partially successful, but usually lost accuracy towards the end. Some candidates clearly knew that the best approach was to find the length BC, but even though this only involved GCSE level maths, were unable to do so.
			Total	13		

Question			Answer/Indicative content	Marks	Guidance
47			$6(1 - \sin^2 x)$ seen eg $6 - 6 \sin^2 x = 5 - \sin x$ $6 \sin^2 x - \sin x - 1 = 0$ $\frac{1}{2}$ and $-\frac{1}{3}$ found $x = \pi/6, 5\pi/6$ [0.52 to 0.524, 2.61799 to 2.62] 3.48 to 3.48143, 5.94 to 5.9435	M1 A1 B1 B2 [5]	at least one correct intermediate step to obtain given answer both required; allow -0.33 or better B1 for 2 correct, to 2 dp or more if B0 allow SC1 for all four answers in degrees with no extras: 30, 150, 340.5 – 341, 199 – 199.5 if B2 deduct 1 mark for extra values in range; ignore extra values outside range Examiner's Comments This was done well by most candidates. A few slipped up with the first part, making sign or bracket errors, but most went on to find the correct values of $\sin x$. Nearly all worked with radians and found $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ successfully. Some gave the other two values in terms of π and lost accuracy, and a small number of candidates decided that the values associated with $\sin^{-1}(-\frac{1}{3})$ had to be outside the range.
			Total	5	

Question			Answer/Indicative content	Marks	Guidance	
48			DR $\frac{dx}{dt} = -\sin t - 3$ $\frac{dy}{dt} = 3 + 4\sin t - 2\cos 2t$ $\frac{dy}{dx} = \frac{3 + 4\sin t - 2\cos 2t}{-\sin t - 3}$ denominator is always negative since $-1 \leq \sin t \leq 1$ Substitution of $\cos 2t = 1 - 2\sin^2 t$ in numerator $4\sin^2 t + 4\sin t + 1$ seen in numerator Numerator is $(2\sin t + 1)^2$ so numerator is always positive and denominator is always negative, so $\frac{dy}{dx}$ is always negative	B1(AO 3.1a) B1(AO 1.1) M1(AO 2.1) E1(AO 2.4) M1(AO 1.1) A1(AO 1.1) E1(AO 2.4) [7]		
			Total	7		
49			Use of cosine rule $\cos \theta = \frac{47^2 + 53^2 - 94^2}{2 \times 47 \times 53}$ 140° or 2.44 radians	M1(AO 3.1) A1(AO 1.1) A1(AO 1.1) [3]	allow if lengths incorrectly placed NB – 0.76635889... allow 140.0, 140.03 or 140.028 or 2.444, 2.4440 or 2.44395	NB 21.23606 or 0.370(6393...); or 18.73589 or 0.327(0030...)
			Total	3		

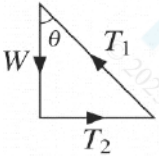
Question			Answer/Indicative content	Marks	Guidance	
50		a	$\tan 45^\circ = 1$ and $\tan 60^\circ = \sqrt{3}$	B1(AO 1.2) [1]		
		b	$\tan(60 - 45) = \frac{\tan 60 - \tan 45}{1 + \tan 60 \times \tan 45}$ $\frac{\sqrt{3} - 1}{1 + \sqrt{3}}$ Multiply numerator and denominator by $\sqrt{3} - 1$ eg $\frac{3 - 2\sqrt{3} + 1}{3 - 1}$ $= 2 - \sqrt{3}$	M1(AO 3.1a) M1(AO 1.1) M1(AO 1.1) A1(AO 2.1) [4]	DR Substitution of their surds in correct compound angle formula AG Convincing arithmetic to given result	Other correct methods eg use of double angle formula are acceptable
			Total	5		

Question			Answer/Indicative content	Marks	Guidance	
51		a	$2\cos\theta + 3\sin\theta \equiv R\sin(\theta + \alpha) \Rightarrow R\cos\alpha = 3,$ $R\sin\alpha = 2$ $\text{So } R^2 = 13 \Rightarrow R = \sqrt{13}$ $\text{and } \tan\alpha = \frac{2}{3}$ $\Rightarrow \alpha = 0.588$	M1(AO 1.1a) B1(AO 1.1) M1(AO 1.1) A1(AO 1.1) [4]		
		b	$k + 2\cos x + 3\sin x > 0$ [for all x] $\sqrt{13}\sin(x + 0.588) + k > 0$ Minimum value of LHS is $k - \sqrt{13}$ $k > \sqrt{13}$	B1(AO 3.1a) M1(AO 1.1) M1(AO 3.1a) A1(AO 2.2a) [4]	oe Use of expression from part (a) Attempt to find minimum value	May be by calculus
		c	$k + 2\cos x + 3\sin x > 0 \Rightarrow \frac{1}{k + 2\cos x + 3\sin x} > 0$	E1(AO 2.4) [1]	oe; accept e.g. statement that the reciprocal of a positive number is positive	
			Total	9		

Question			Answer/Indicative content	Marks	Guidance	
52		a	$a = 40$ $c = 180$ $30 = 40 - b(315 - 180)^2 \Rightarrow b = \frac{2}{405}$ AG	B1(AO 3.3) B1(AO 3.3) B1(AO 2.1) [3]	At least one intermediate step needed	
		b	(A) 15.8	B1(AO 3.4) [1]	Allow integer answers 15 or 16	
		b	(B) 250 is outside the interval which provides the data and so may be extrapolating too far	E1(AO 3.5a) [1]		
		c	(A) $t < 90 \Rightarrow N < 0$ which is not meaningful	E1(AO 2.4a) [1]		
		c	(B) $t > 270$	E1(AO 3.5b) [1]		
		d	Reflection in t -axis or translated by $\begin{pmatrix} 180 \\ 0 \end{pmatrix}$ or $\begin{pmatrix} -180 \\ 0 \end{pmatrix}$ Stretch in the y -direction, scale factor 40	B1(AO 1.1a) B1(AO 1.1a) [2]	Vector notation required for translation	Accept stretch in the y -direction, scale factor -40, for B2
		e	$N = -40\cos t$ or $N = 40\cos(t - 180)$	B1(AO 3.3) [1]	oe, e.g. $N = 40\cos(t + 180)$	

Question			Answer/Indicative content	Marks	Guidance
		f	(A) Stretch graph in t -direction by scale factor $\frac{365}{360}$ or modify equation to $N = -40 \cos\left(\frac{360}{365}t\right)$	B1(AO 3.5c) [1]	oe, e.g. $N = 40 \cos\left(\frac{360}{365}t - 180\right)$; FT from their (v) for modified equations
		f	(B) Maximum point moves to (182.5, 40) or as consistent with their answer to (A)	B1(AO 3.4) [1]	N.B: $N = 40 \cos\left(\frac{360}{365}(t - 180)\right)$ oe will 365 leave maximum point unchanged
			Total	12	

Question			Answer/Indicative content	Marks	Guidance	
53		a	$\sec \theta - \cos \theta \equiv \frac{1}{\cos \theta} - \cos \theta$ $\equiv \frac{1 - \cos^2 \theta}{\cos \theta}$ $\equiv \frac{\sin^2 \theta}{\cos \theta} \equiv \frac{\sin \theta}{\cos \theta} \sin \theta$ $\equiv \tan \theta \sin \theta \text{ AG}$	B1(AO 1.2) M1(AO 2.1) M1(AO 1.2) E1(AO 2.1) [4]	Use of definition of sec Manipulation of fractions Use of at least one trig identity Clear and complete argument	
		b	$\tan \theta \sin \theta = \frac{1}{2} \tan \theta \Rightarrow \tan \theta \left(\sin \theta - \frac{1}{2} \right) = 0$ $\tan \theta = 0$ gives $\theta = 0, \pi$ $\sin \theta = \frac{1}{2}$ gives $\theta = \frac{1}{6}\pi, \frac{5}{6}\pi$ Alternative method $1 - \cos^2 \theta = \frac{1}{2} \sin \theta \Rightarrow \sin^2 \theta = \frac{1}{2} \sin \theta$ $\sin \theta = 0$, gives $\theta = 0, \pi$ $\sin \theta = \frac{1}{2}$ gives $\theta = \frac{1}{6}\pi, \frac{5}{6}\pi$	M1(AO 1.1a) A1(AO 1.1b) M1(AO 1.1a) A1(AO 1.1b) M1 A1 M1 A1 [4]	Factorising Both values At least one value from arcsin Both values correct Obtaining quadratic in $\sin \theta$ Both values At least one value from arcsin Both values correct	SCI for sole answer $\frac{1}{6}\pi$ www

Question			Answer/Indicative content	Marks	Guidance
			Total	8	
54			T_1 and T_2 are tensions in wire and string respectively; the globe has mass m Resolve vertically: $T_1 \cos \theta = mg$ $T_1 = \frac{mg}{\cos \theta}$ Resolve horizontally: $T_2 = T_1 \sin \theta$ $T_2 = \frac{mg}{\cos \theta} \times \sin \theta = mg \tan \theta$ $mg \tan \theta < mg$ for $\tan \theta < 1$ $\theta < 45^\circ$ Alternative method  Triangle of forces Maximum T_2 is when $T_2 = W$ In this case $\theta = \arctan(1)$ Hence $\theta < 45^\circ$	M1(AO 3.1b) M1(AO 3.3) M1(AO 2.1) A1(AO 2.2a) M1 M1 M1 A1 [4]	Allow sin/cos interchange for 1st M1 only Use of trig identity Inequality clearly stated Forces shown, correct arrow directions Equating T_2 and weight for limiting case Use of trigonometry or isosceles triangle Inequality clearly stated
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
55		a	Area of triangle $= \frac{1}{2} \times 10^2 \sin \frac{1}{6} \pi$ ($= 25$) $AC = 2 \times 10 \sin \frac{1}{12} \pi$ ($= 5.17638$) Area of semicircle $= \frac{1}{2} \pi \times \left(\frac{5.17638}{2} \right)^2$ ($= 10.5223$) Total area $= 35.5 \text{ cm}^2$	B1(AO 1.1a) M1(AO 3.1a) M1(AO 1.1b) A1(AO 1.1a) [4]	Correct unsimplified expression so Or use of cosine rule Use of $\frac{1}{2} \pi \times$ (half of their AC)²
		b	$50\theta = 35.52$ $\theta = 0.710$ to 3sf	M1(AO 1.1a) A1(AO 1.1b) [2]	Equating $\frac{1}{2} r^2 \theta$ to their answer from (a)
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
56			$v = \frac{ds}{dt} = 3 + 0.3t^2$ and $a = \frac{dv}{dt} = 0.3 \times 2t$ $a = 0.6t$ Acceleration is a function of t so is not constant Alternative method For constant acceleration, $s = ut + \frac{1}{2}at^2 (+c)$ i.e. s is a quadratic function of t So this cubic function is not constant acceleration	M1(AO 1.1a) A1(AO 2.1) E1(AO 2.2a) M1 A1 E1 [3]	Attempt to differentiate twice Must be clearly argued Comparing with standard formula Explicit identification of quadratic, oe Deduction clearly stated
			Total	3	
57			$2\theta = \cos^{-1} 0.3$ $2\theta = 72.54, 287.46, 432.54, 647.46$ $\theta = 36.3, 143.7, 216.3, 323.7$	M1(AO 1.1a) A1(AO 1.1b) A1(AO 1.1b) [3]	For at least one correct answer in the range For all correct with no others in the range
			Total	3	